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Advances in solar photovoltaic tracking systems: A review

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ABSTRACT

Solar photovoltaic technology is one of the most important resources of renewable energy. However, the current solar photovoltaic systems have significant drawbacks, such as high costs compared to fossil fuel energy resources, low efficiency, and intermittency. Capturing maximum energy from the sun by using photovoltaic systems is challenging. Several factors that affect the energy output of such systems include the photovoltaic material, geographical location of solar irradiances, ambient temperature and weather, angle of sun incidence, and orientation of the panel. This study reviews the principles and mechanisms of photovoltaic tracking systems to determine the best panel orientation. The tracking techniques, efficiency, performance, advantages, and disadvantages of simple tracking systems are compared with those of state-of-the-art tracking systems. Diverse types of solar tracking systems based on their technologies and driving methods will be presented and categorized.The future trends of tracking systems are also highlighted.

1. Introduction

Numerous researchers worldwide have increased their attention to the development of renewable energy. Renewable energy sources are friendlier to the environment and more sustainable compared to fossil fuels [2–6]. Renewable energy includes several forms such as solar and wind energy. The interest in using renewable energy instead of fossil fuel has increased, and several countries have increased their dependence on solar energy. The total amount of consumed solar energy worldwide in million tones oil equivalent is shown in Fig. 1.

As shown in Fig. 1, the total amount of solar energy that consumed worldwide increased exponentially from 2000 to 2016. The maximum amount of consumed solar energy was observed in 2016 whereas solar energy was slightly used before this year. Official data from various government ministries and statistical offices were used to compile these data [13]. The total gross generation of solar energy worldwide in Terawatt-hours is shown in Fig. 2, while Fig. 3 shows the total capacity in Megawatt.

As shown in Figs. 2 and 3, the total capacity, generated, and consumed energy has increased exponentially, and the total growth of solar energy capacity and usage is 29.6%.

Solar photovoltaic cells or solar panels have been used for decades to convert solar energy into electricity. Solar photovoltaic cells are a scalable technology depending on the size of the load. Photovoltaic cells can be used to power small electronics or can be wired together to make solar panels for larger size loads [14–16]. The panels can be collected together to create a solar array for large-scale power generation [9,17].

A single diode model is commonly used to present the solar photovoltaic cells [18–20]. The equivalent circuit for photovoltaic cell is shown in Fig. 4.

Fig. 4 represents that the solar photovoltaic cell is modeled by using a current source that generated depending on the solar radiation in parallel with a diode. Shunt resistance and series resistance are used to obtain the solar photovoltaic cell output. The output current can be calculated as follows.

$$I = I_{ph} - I_0\left(\exp\left(\frac{V + IR_s}{N_s V_{TH}}\right) - 1\right) - \frac{V + IR_s}{R_p} \quad (1)$$

where, V is the module voltage, I_0 is the diode current, N_s is the number of the series cells in the module, V_{TH} is the PN junction thermal voltage that can be calculated as follows.

$$V_{TH} = \frac{k(T_c + 273)}{q} \quad (2)$$

where, k is the Boltzmann constant $k = 1.38\times10^{-23} j/K$, q is the electoral load that $q = 1.6\times10^{-19} C$, and T_c is the temperature in Celsius.

Solar tracker systems are designed and developed to increase the amount of solar radiation received by photovoltaic devices. This process is carried out by maintaining the optimum angle of the solar panel to produce the best power output [21,22]. Solar tracking systems have been used in numerous places worldwide. Many solar tracking systems have been built and designed to achieve the optimal amount of solar energy, and many models have been proposed to enhance the

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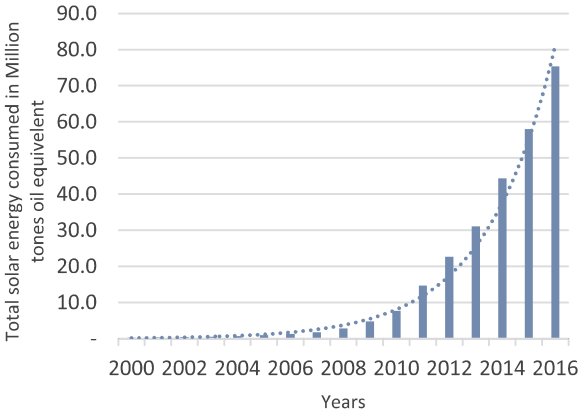


Fig. 1. Total solar energy consumed in Million tones oil equivalent.

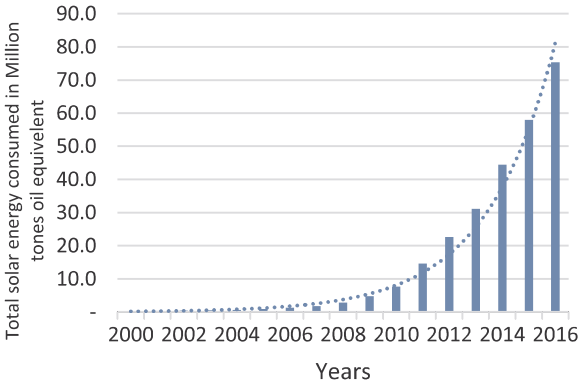


Fig. 2. Total solar energy consumed in Million tones oil equivalent.

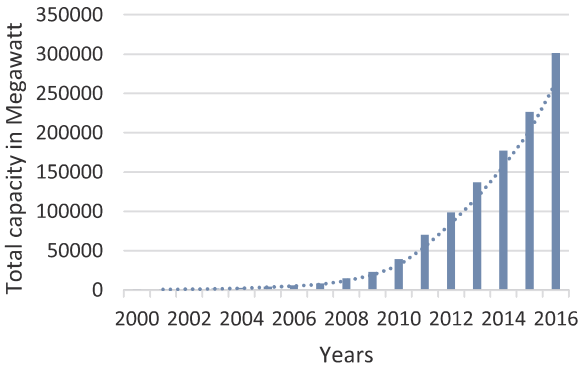


Fig. 3. Total solar energy capacity in Megawatt.

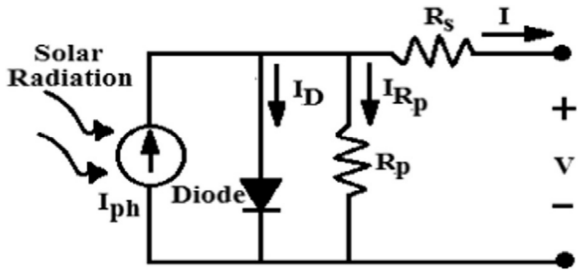


Fig. 4. Equivalent circuit for solar photovoltaic cell.

advantages of solar panel usage. Several studies have focused on designing and implementing solar tracking systems for different geographical regions [23–34]. Fig. 5 shows the difference between using a fixed-angle solar photovoltaic and a simple tracking system.

Tracker systems track the position of the sun, thereby increasing the input of solar radiation and electrical energy output [32,35,36]. However, designing, implementing, and installing these systems are difficult for different reasons. Multiple amount of measurement results are required before employing tracker systems [37–39]. These results are collected during a relatively long period time to be used when installing solar cells to track the sun [29]. The collected results are used to identify the best technique for tracking the position of the sun. Following the position of the sun is performed to obtain the optimal output of solar energy in all situations. Different environmental pressures and different parameters, including panel direction, angle of photons incidence, time to measure the results, material of solar cells, and conductivity of photovoltaic modules, may affect the output of the solar panel cells [40–42].

This study seeks to present and discuss diverse types of solar tracking systems. The discussion is based on the principles and technologies that used to implement and drive these systems. Passive and active solar tracking systems are presented, and a comparison among the control methods used to drive solar tracking systems is conducted. This study is to categorize the solar tracking systems based on their control methods. Different principles are presented in a chronological order: from passive trackers to tracking systems that employ the artificial intelligence (AI).

Section 2 discuss solar tracking systems and a few important parameters for their installation. Their performance is also evaluated. Two main types of solar tracking systems, namely, passive are discussed in Section 3. Section 4 presents active solar tracking systems that can be classified into numerous categories depending on their driving methods. The limitations and the future trends of solar tracking systems are provided in Section 5. Finally, the conclusion is presented in Section 6.

2. Solar tracking systems

A solar tracking system tracks the position of the sun and maintains the solar photovoltaic modules at an angle that produces the best power output. Several solar tracking principles and techniques have been proposed to track the sun efficiently. The idea behind designing a solar tracking system is to fix solar photovoltaic modules in a position that can track the motion of the sun across the sky to capture the maximum amount of sunlight. Tracker system should be placed in a position that can receive the best angle of incidence to maximize the electrical energy output. Designing such device to produce electrical energy is interesting and important. However, it requires extensive mathematical calculations and detailed measurements of different solar parameters. One of the most important parameters is the daily average solar irradiance. The daily average solar irradiance ranges from 4–7 kilo-Watt-Hour (KWh/m^2) worldwide [43]. This high amount of daily solar irradiance has led to the use of solar energy for electricity generation in several applications, such as water pumping, telecommunication, and lightening. Sun irradiance varies with months, time of day, weather conditions, geographical area, and position of the sun in the sky [44].

Several solar parameters must be considered when designing, implementing, and installing tracker systems. Greenwich Time and Solar Time are the basic parameters in the solar energy field. Greenwich Time is a timescale that depends on the rotation of the earth around itself in one day [45–47]. This time can be determined based on the meridian that passes through the town of Greenwich in Britain, which is one of the longitudes that was adopted as a reference point for world timing. Other longitudes are divided into two parts based on their positions along the Greenwich line. The time in all east lines is calculated by adding a number of hours (+) to the Greenwich Time. The time in all west lines is calculated by subtracting a number of hours (-) from the Greenwich Time [48,49]. The number that is added or subtracted to each Greenwich Time is determined based on the region of the time zone relative to the Greenwich line. A total of 360 longitudes exist all

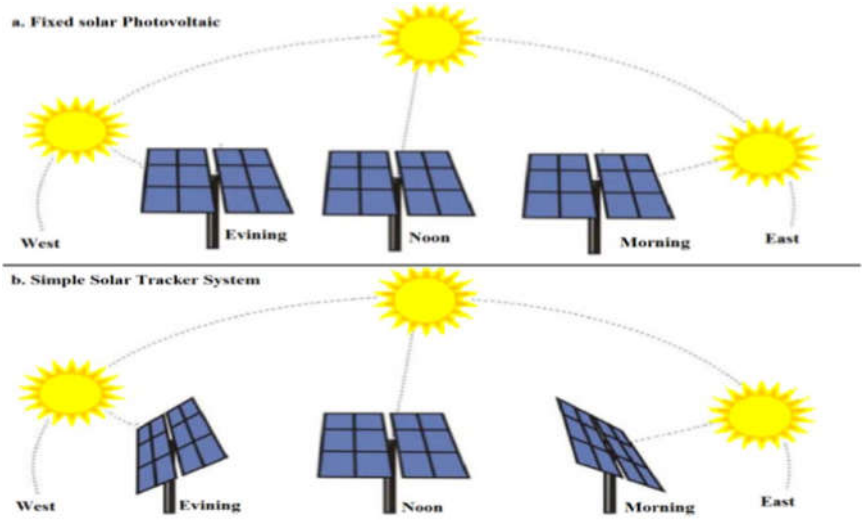


Fig. 5. Fixed angle solar panel (a) and solar panels with a tracking system (b).

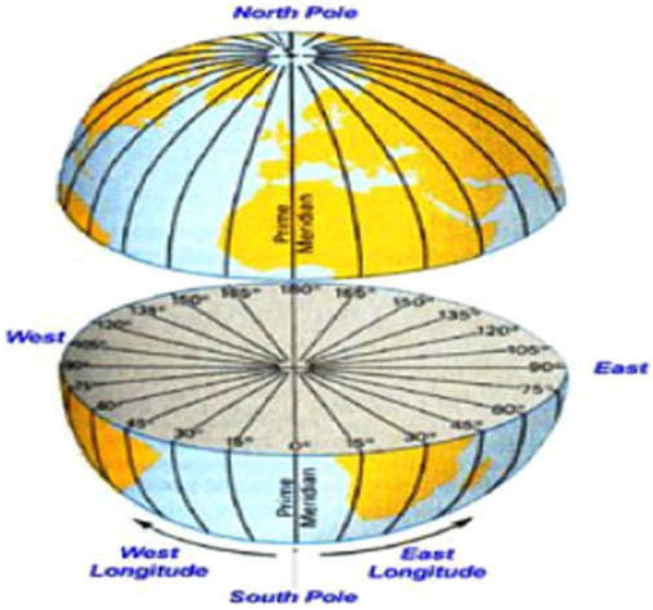


Fig. 6. Longitudes of the earth [45].

over the world based on the Greenwich line and each time zone, as shown in Fig. 6. A variation on longitudes can occur based on the movement of the earth. The earth can move by one degree every four minutes, and it rotates around itself once per day [50]

The Solar Time parameter is also important because it is the real time that depends on the position of the sun in the sky. The fundamental unit in this time is the day order in the year starting from 1 on the 1st of January and ending with 365 on the 31st of December [51].

Solar Time can be calculated as follows.

$$SolarTime = StandardTime + 4(L_{ST} - L_{Loc}) + EoT \quad (3)$$

where, Standard Time is the local time of a region, L_{st} is the standard longitude that is positive for east regions and negative for west, L_{Loc} is the local standard meridian time, and EoT is the equation of time.

The local standard meridian time can be calculated by multiplying the time zone of a region by 15° per hour as follows.

$$L_{Loc} = Timezone \times 15° \quad (4)$$

The equation of time is the time difference between the apparent solar time for people and the real mean solar time. The EoT can be calculated as follows.

$$EoT = 229.2(0.000075 + 0.001868\cos B - 0.032077\sin B - 0.014615\cos 2B - 0.04089\sin 2B) \quad (5)$$

where, β is the displacement of time from standard time such that:

$$B = \frac{(n-1) \times 360}{365} \quad (6)$$

where, n is the day order in the year.

In addition to determining the longitude and equation of time, a few parameters should be calculated or measured to determine the best direction and orientation of solar photovoltaics modules, therefore, obtain more output power. Latitude, angle of incidence, light intensity, tilt angle, orientation angle, solar azimuth angle, declination angle, inclination angle, elevation angle, and zenith angle are the main parameters that specify the best location and direction of solar tracking systems. Latitude is a parameter used to determine the angular distance (south or north) of the equator in any location on Earth. The latitude angle is measured in degrees, as shown in Fig. 7.

Angle of incidence is the most important parameter in installing solar tracking systems. Angle of incidence is the angle between the rays of the sun falling on the surface and the rays perpendicular to that surface [48], as shown in Fig. 8, where the angle of incidence is indexed by (θ°).

However, using the angle of incidence alone is insufficient when installing solar tracking systems. Light intensity, which is commonly

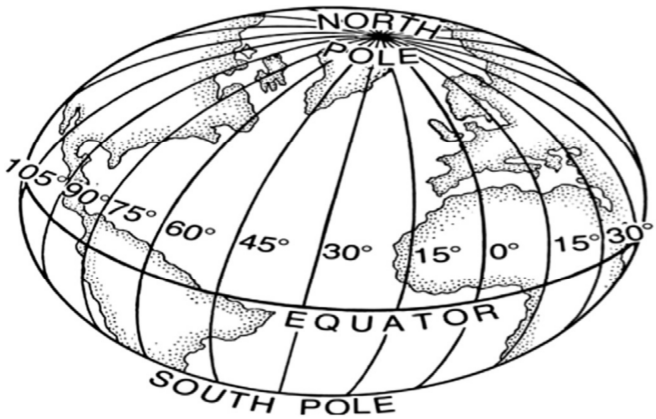


Fig. 7. Equator line and latitude [45].

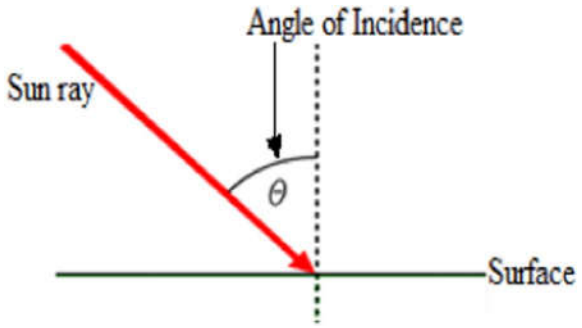


Fig. 8. Angle of incidence.

called solar irradiance of a light source, is also an important parameter to install tracking systems. Light intensity can be determined by measuring measuring either the power of the light source or the luminous flux. The light intensity of the sun is measured using specific tools, such as a Pyranometer.

Several other parameters have to be measured to sufficiently install the solar tracking systems. Declination angle, which is the angle between the equator and a line drawn from the center of the sun to the center of the earth, is one of these parameters. The declination angle is indexed by (δ) and is measured by degrees, as shown in Fig. 9.

The declination angle can be calculated as follows.

$$\delta = -23.45 \times \cos\left(\frac{360}{365} \times (n + 10)\right) \quad (7)$$

Elevation and zenith angles have close definitions to the declination angle. Elevation angle is the altitude of the sun or the angle between the center of the sun and the horizon. Zenith angle is the angle between the center of the sun and the vertical. Elevation and zenith angles are measured in degrees as shown in Eqs. 8 and 9, and they are indexed by (α) and (z), respectively. Fig. 10 shows both elevation and zenith angles.

$$\alpha = 90° - z \quad (8)$$

$$z = \cos^{-1} \sin L_{st} \sin\delta + \cos L_{st} \cos\delta \cos Standard\,Time \quad (9)$$

Tilt and orientation angles are very important parameters for sufficiently tracking the sun. Tilt angle is the angle between the solar tracking system and the horizontal axis. Angle of incidence is a form of the tilt angle [52]. Orientation angle can be used to move a solar tracking system to keep the solar photovoltaic module perpendicular to the sun in order to obtain optimum power generation, as shown in Fig. 11.

Solar azimuth angle is measured with respect to due south, it is the angular distance between the projection of the line of sight to the sun on the ground and due south. Solar azimuth angle is indexed by (γ_s) and is indicated by a positive sign for the position east of south or a negative sign for the position west of south. Azimuth angle is shown in Fig. 12.

Solar azimuth angle can be calculated using Eq. 10 as follows.

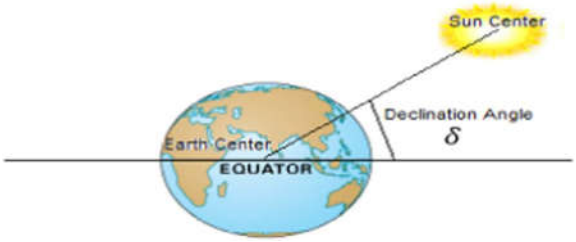


Fig. 9. Declination angle.

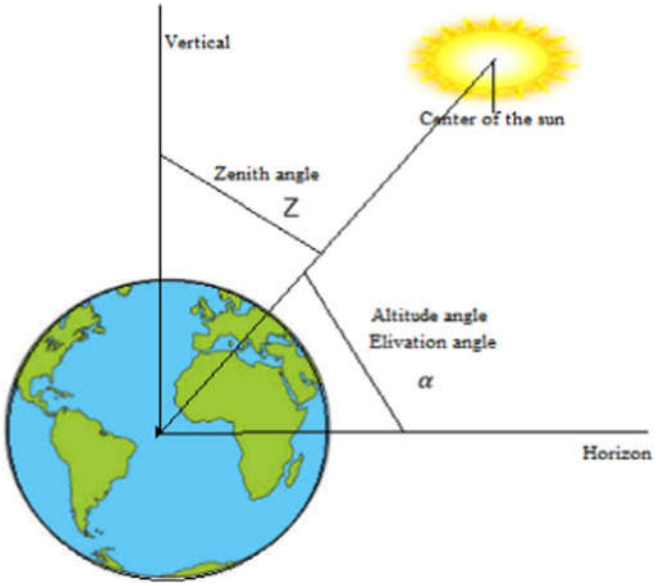


Fig. 10. Elevation and zenith angles.

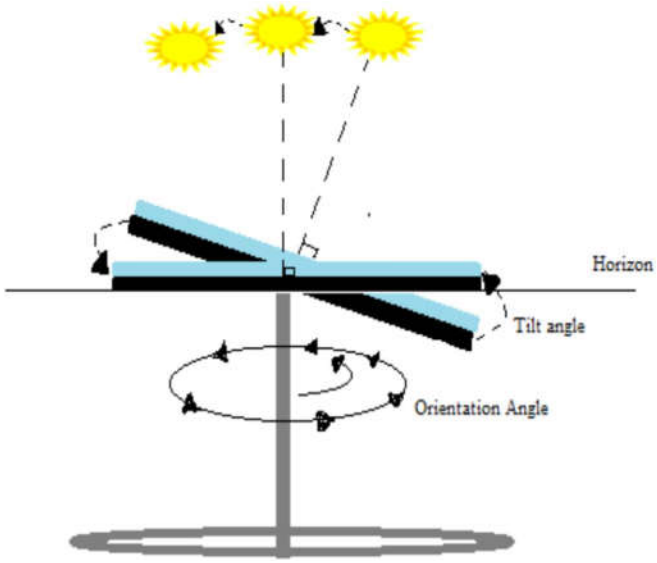


Fig. 11. Tilt and orientation angles.

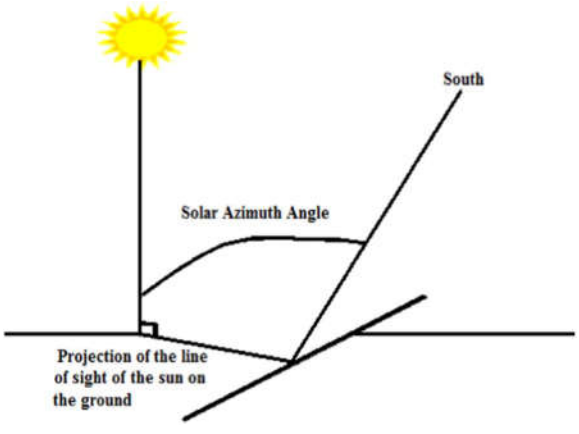


Fig. 12. Solar azimuth angle.

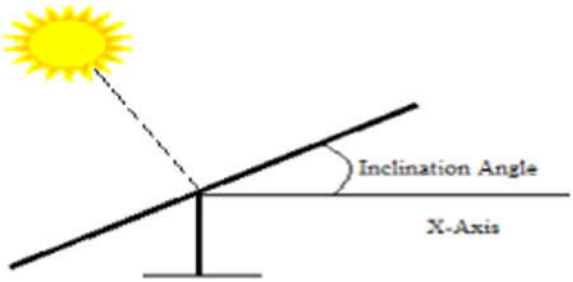


Fig. 13. Inclination angle.

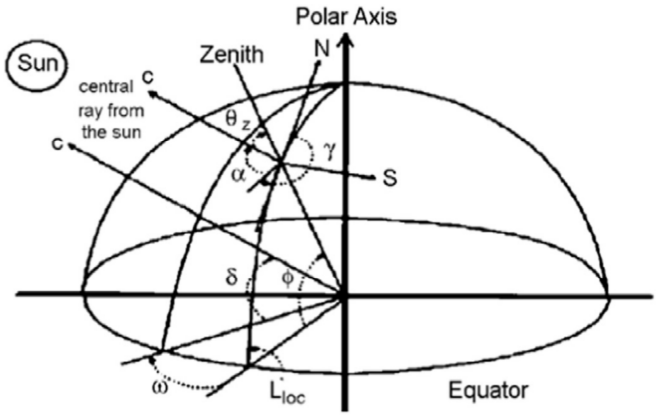


Fig. 14. Schematic representation of the solar angles [53].

$$\gamma_s = \sin^{-1}\left(\frac{-\sin h \cos\delta}{\sin z}\right) \quad (10)$$

where, h is the hour angle that

$$h = 15°(solartime - 12) \quad (11)$$

Inclination angle is the angle between a photovoltaic module and the positive x-axis. This angle is indexed by (β) and it measured in degrees, as shown in Fig. 13.

All of the preceding parameters can be utilized to determine the best location to install solar tracking systems or find the best orientation for such systems to obtain higher power, as shown in Fig. 14.

The complexity of a tracking system depends on the number of axes used to move the solar photovoltaic modules horizontally, vertically, or both. Two main types of solar tracking systems exist. The first one is single axis tracking, which can be used to move the solar photovoltaic horizontally or vertically. The second type is dual axis solar tracking systems, also known as two-axis tracking, which can be used to simultaneously change in both directions. Work in the field of solar tracking systems does not stop after installing a photovoltaic panel in a suitable place. Solar tracking systems should be evaluated to ensure that they function efficiently and that they can be used in the future of solar energy. Numerous evaluation metrics are utilized to evaluate solar tracking systems. The type of driver or control system is one of the metrics used to determine the next direction of the photovoltaic panels and the axis angle movement. This operation is complex because the idea is to direct the photovoltaic panels to the optimal direction to generate higher solar energy. The testing method is another metric used to evaluate solar tracking systems. Solar tracking systems can be implemented using hardware or a combination of hardware and software. Numerous solar tracking systems have been proposed around the world, and many of them have been implemented. Others have only been simulated or theoretically proposed as an idea.

Moving solar tracking systems from side to side is also important point. Solar tracking systems can be manually moved mechanically through the use of cantilevers, gears or motors. The most important metric to assist the proposed solar tracking systems is calculating the gained energy compared with the consumed energy by the tracker components. Motors, hardware components, resistors, and the size of photovoltaic panels can affect the gained power. The difference between real and expected results is also a an important point. This difference is called the error rate, which can evaluate whether the system passes the test or should be upgraded. The cost to practically implementing a solar tracking system is acrucial evaluation metric for the commercial implementation of solar tracking systems. Costs are presented in numerous forms, such as the system complexity, energy consumption of the tracking system, and error rate or accuracy of the tracking system.

3. Types of solar tracking systems

Solar tracking systems can be mainly divided into two main groups based on the techniques that control the photovoltaic module [32]. These two main groups are active and passive tracking system. Active tracking systems use motors and gear trains to direct the panel toward the sun. Passive tracking systems use a low boiling point compressed gas fluid that originates from solar heat. Passive and active solar tracking are the two main techniques utilized to efficiently track the sun [25,54–58].

3.1. Passive solar tracking systems

Passive solar tracking systems use the temperature of the sun to heat a kind of liquid which is stored in canisters and is affixed to the sides of photovoltaic modules. The idea is to heat up this kind of liquid in the canisters and expand it into a gas. The expanded gas can push the heavier liquid to the shaded canister and shift the weight to that side of the photovoltaic module, thus creating a gas pressure that causes it to move and rotate [59]. Fig. 15 shows the working process of passive solar tracking systems.

The first commercial passive solar tracking system was produced by Zomeworks Corporation in 1969. Zomeworks Corporation has been the leader in passive solar energy products since 1969. The Universal Track Rack™ series are the most popular products of Zomeworks for large arrays. Since 1980, more than 17,000 Zomeworks tracking systems have been installed in different locations and climates all over the world [60].

In Zomeworks Track Racks, the electrical output of photovoltaic modules is increased by 25% compared with that of modules on fixed mounts. Track Racks systems are highly cost-effective components for industrial power systems, water pumping systems, cathodic protection systems, and utility applications [1,61].

Elmaged et al [7] evaluated the performance of solar pumping systems by installing the solar photovoltaic modules in two different modes. The first is the fixed tilted mode, in which photovoltaic modules are installed facing due to the south. The second mode is the passive solar tracking mode, in which solar photovoltaic modules are mounted on a passive solar tracker. The passive solar tracker is installed facing due to the south and inclined to the horizon. The passive solar tracker can rotate from east to west to follow the apparent daily movement of the sun. The evaluation of both systems showed that using a passive solar tracker can increase the accumulated solar radiation incident by 14% and the accumulated power by 16%.

Narendrasinh et al. [8] proposed, tested, and evaluated a new passive solar tracker system that uses three types of gases. The new passive solar tracker system is concentrated around a solar tracker working in the Zomework principle. The main idea of this tracker system is to propose a passive model that has been implemented and tested using three types of gases: thinner, methanol, and acetone. Evaluation of this passive model showed that Zomework passive solar tracker systems have 23.33% more power than fixed tracking systems.

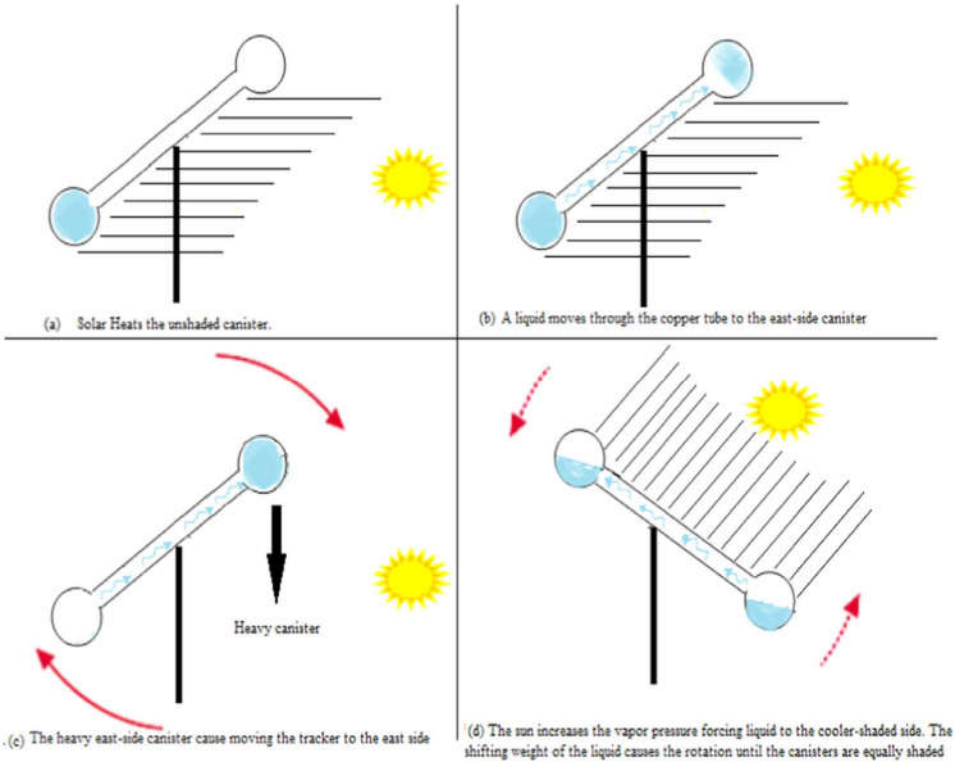


Fig. 15. Passive solar tracking system.

By contrast, Clifford et al. [11] presented a novel passive tracking system activated by aluminum/steel bimetallic strips and controlled by a viscous damper. Two bimetallic strips comprising aluminum and steel were used to design this passive solar tracker and position it on a wooden frame. The designed passive solar tracker was experimentally tested and evaluated, and the measured efficiency increased by 23.3%.

The main advantage of the passive system is its capability to track the sun from side to side without using motors, gears, or controllers. The other advantages of passive systems over active ones include a relatively easy installation process, effective results, absence of external power requirements, and a low maintenance cost. Passive solar tracking systems are still currently used in many places worldwide. However, passive solar tracking systems suffer from numerous problems that affect their performance. Table 1 shows a comparison between different types of passive tracking systems.

The main drawback of passive solar tracking systems is its strong dependency on weather conditions. Although passive solar tracker can maximize heating from the sun, bad weather conditions can render these trackers inefficient. The location of the installation of this type of solar trackers is important because the selected location must receive sufficient sustained sunlight for an efficient solar heating process. Moreover, harsh winters can overwhelm passive solar tracking systems, thus requiring additional assistant devices and increasing costs. Another drawback is selecting the appropriate types of glass and gas to build passive solar tracking systems with increased efficiency. The absorption level of glasses varies from type to type and depends on the chemical properties, strength, and color of glasses. The chemical properties and characteristics of different types of gases vary from type to type as well. Changing the type of gases utilized in passive solar tracking systems can also affect the performance of these trackers. The orientation of passive solar tracking systems is imprecise; thus, it is unsuitable for a few types of concentrated photovoltaic modules. However, they can work effectively for common photovoltaic types or large photovoltaic panels. Numerous factors, especially wind, heavy rain, and clouds, can prevent the excessive motion of passive solar tracking systems [62,63]. The type of active solar tracking system can solve the problems of using passive solar tracking systems.

Table 1
Comparison between different passive solar tracking systems.

Technology	Characteristics	Advantages	Disadvantages
Zomework Track Rack [1]	• Single axis • Hardware implemented • Compared with fixed tracking	• Cheap • Increased the electrical output by 25%	• Easily affected by weather
Solar pumping system [7]	• Single axis • Hardware implemented • Compared with fixed tracking	• Capable of rotating from east to west with the sun movement. • Increased the accumulated solar radiation incident by 14% and accumulated power by 16%	• Easily affected by weather
Zomework Principle [8–10]	• Single axis • Hardware implemented • Compared with fixed tracking	• Uses three types of gases • Powerful • Increased the power by 23.33%	• Expensive • Easily affected by weather
Novel Passive [11,12]	• Single axis • Hardware implemented • Compared with fixed tracking	• Cheap • Increased the efficiency by 23.3%.	• Easily affected by weather

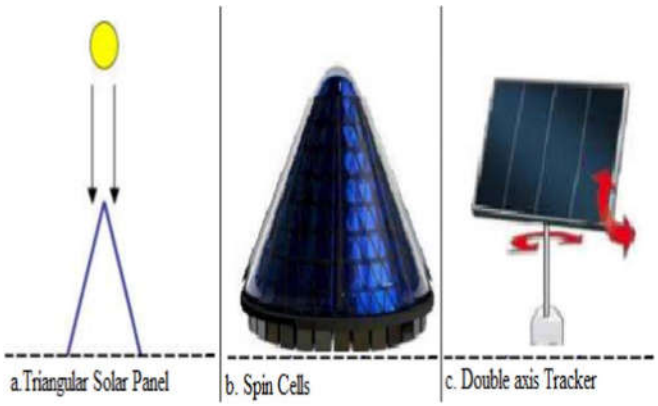


Fig. 16. Triangular solar tracker, spin cell, and double solar tracker [64].

3.2. Active solar tracking systems

Active solar tracking systems are systems that use motors, gears, and other controllers to direct the photovoltaic panels toward the sun. Active tracker systems come in several varieties that can be classified into a few categories. Aman et al. [64] classified active solar tracking into four categories, namely, triangular solar panel, single axis tracking, double-axis tracking, and spin cell, as shown in Fig. 16. The triangular tracking system uses two solar photovoltaic modules facing opposite directions, and both modules can receive equal amounts of sunlight. The single axis tracking system is the simplest and cheapest tracker; however, its effectiveness is low because the photovoltaic module can be directed either horizontally or vertically only. Meanwhile, in the double tracking system, the photovoltaic module can be directed horizontally and vertically. In spin cell solar tracking systems, solar cells are placed on a cone frame. Moving the photovoltaic modules is unnecessary in this tracking system because it is efficient and it can obtain high efficiency, thereby increasing the measured current per cell by 20 times compared to the flat modules [64].

Otieno et al. [65] divided active solar tracking systems into three categories: chronological solar tracking, single axis trackers, and dual axis trackers. Chronological tracking is designed to follow the movement of the earth using the same direction and speed as those of the earth. Elmaged et al. [7] divided all types of solar tracker systems into two categories: general active and passive trackers. Mousazadeh et al. [66] and Loschi et al. [67] classified active solar tracking systems into four categories based on the technologies utilized to change the directions of photovoltaic modules. The categories are based on control as microprocessor and electro-optical-based, auxiliary-bifacial-cell-based, date-and-time-based, or a combination of these categories. Microprocessor and electric-optical sensor trackers can control the photovoltaic modules based on a feedback mechanism. The number of inputs is driven from sensors and other electronic devices that detect relevant parameters of the sun and send them to a processor that can analyze these parameters to find a suitable output.

Auxiliary bifacial cell-based tracking systems use double-face photovoltaic modules that can be directly connected to a magnet installed on a DC motor that is already located at the rotary, as shown in Fig. 17. The system measures the values of solar irradiation and electric power supply to track the sun. In date-and-time-based tracking systems, the results of the system are not monitored; therefore, the system cannot correct errors. Date and time controllers only depend on calculating the position of the sun using numerous formulas, algorithms, and mathematical equations. The calculated position of the sun is combined with geographical and astronomical information to calculate the value that drives the photovoltaic modules.

The number of axes that is commonly used to move the photovoltaic modules is the most important factor for all solar trackers in the world.

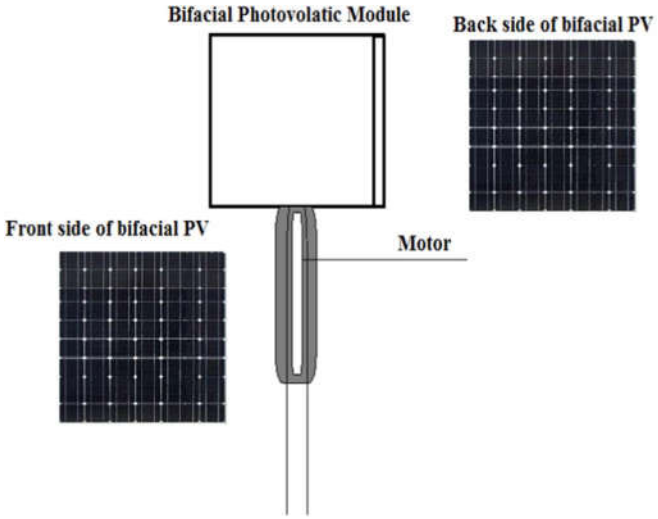


Fig. 17. Bifacial photovoltaic solar tracking system.

Single and dual-axis tracking systems are the two main types used. The single-axis tracking system, wherein the solar panel can be moved only in one of two directions (either horizontally or vertically), is the simplest tracker [33]. Other classifications of solar tracking systems have been proposed based on the movement of solar photovoltaic modules. Solar tracking systems have been classified as fixed and movement solar tracking systems. In fixed solar tracking systems, the solar photovoltaic modules can be directed toward a specific direction for a long time in a year, and a few changes can be applied depends on the weather situations. In movement solar trackers, the solar photovoltaic modules can be controlled to follow the position of the sun for the entire year and the entire day. The fixed tracking system is cheaper and simpler than the movement tracker; however, it is also less efficient and gains less power. Active solar tracking systems can also be classified based on the driving methods or control logic they adopt. In addition to manual controllers, numerous driver systems can be utilized to control the solar tracking systems.

4. Solar tracking systems based on driving methods

Single-axis tracking systems are nearly the best solution for small photovoltaic modules. These systems usually involve manual elevation or axis tilt adjustment on the second axis, which is adjusted at regular intervals throughout the year. The single-axis system is cheaper than other systems, and it can be moved horizontally or vertically depending on the location of the sun and weather.

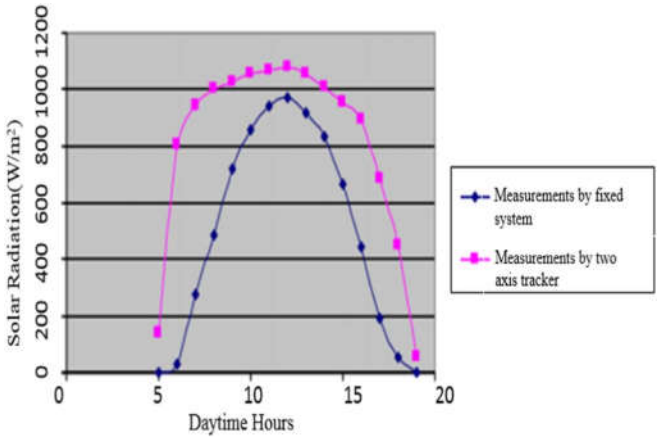


Fig. 18. Solar radiation measurements [23].

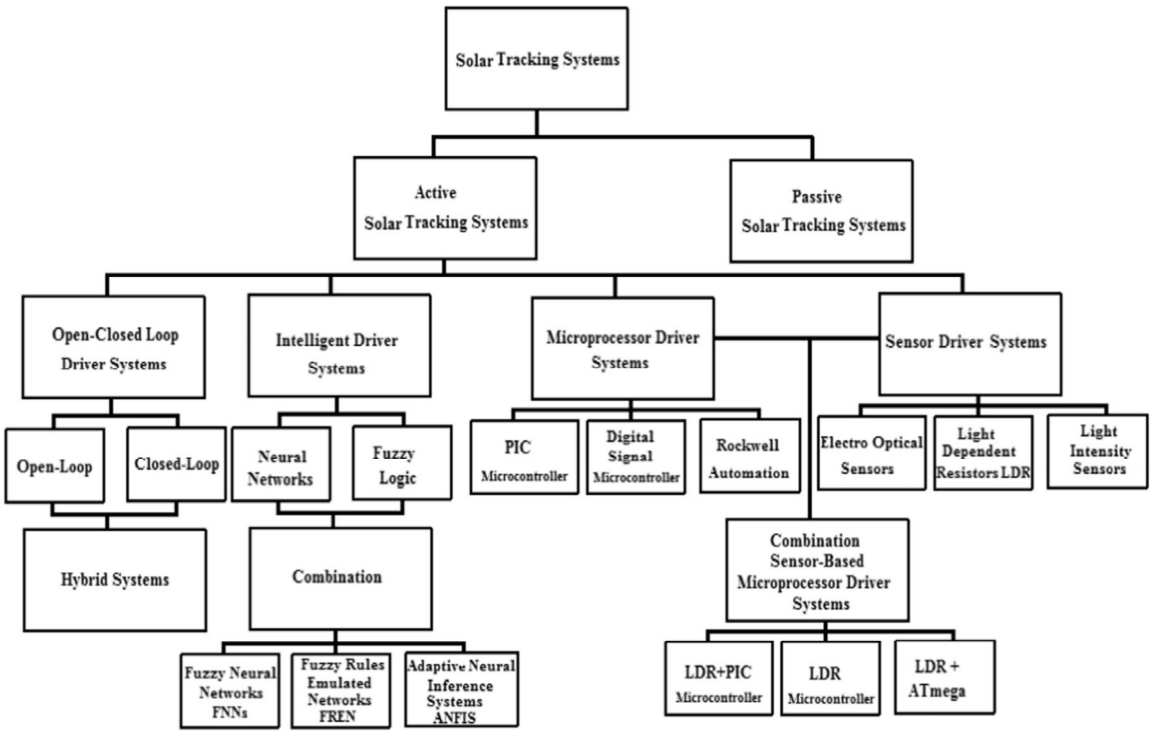


Fig. 19. Solar tracking systems classification based on driving methods.

This study categorizes tracking systems based on their controlling methods. Controlling methods are the methods used to move the solar tracking systems from side to side. Controlling methods, which are also known as driving methods, can be divided into five types: sensor driver systems, microprocessor driver systems, open-closed loop driver systems, intelligent driver systems, and a combination of two or more of these driver systems. Fig. 19 shows the suggested classification and several examples of the procedures used for each type.

4.1. Sensor driver systems

Photo sensors, which are also known as photo detectors, are sensors of light that can detect light resources and convert the light signals into voltage or current. Photodiodes and phototransistors are a few examples of photo detectors. Solar cells are similar to photodetectors because they absorb light and turn it into energy. Electro-optical sensors are electronic detectors that can detect a change in light and convert it into an electronic signal that can be used in numerous industrial and consumer applications. Optical sensors can measure the physical quantity of light and translate it into a readable form by an instrument using electrical triggers. This trigger is the part that reacts to the change in light when the optical sensor measures the changes from a light beam, thereby increasing or decreasing the electrical output.

Numerous types of optical sensors have been utilized to drive the solar tracking systems. Light Dependent Resistors (LDRs), light optical sensors, and light intensity sensors are a few examples of optical sensors used to drive solar trackers and change their directions to follow the position of the sun.

Zogbi et al. [68] designed and constructed a reliable and low-cost two-axis sun tracker system. This sun tracking system was fastened on a universal two-axis frame to provide rotation in both azimuth and elevation. The proposed sun tracker is controlled by a classical electronic system containing four electro-optical sensors, two motors, a differential amplifier, and other electronic components. The four electro-optical sensors are placed in each quadrant to provide a complete view of the sky. The idea is to feed the output of each pair of sensors to the differential amplifier, which can compute both inputs to find a DC error signal. This DC signal can be amplified to drive the corresponding steering motor until a zero output is obtained from the differential amplifier.

Assaf [23] designed and implemented a two-axis solar tracking system that uses the propagation of the sun beam to measure solar radiation. Four LDRs, two servo motors, and a programmable logic controller (PLC) device were used to build and test this solar tracking system. The LDRs track the position of the sun and provide a voltage signal to the PLC. The PLC contains a software that can use the information of the sensors to generate an output signal to the motors. The idea is to develop a control software to determine the optimum position of the photovoltaic panel during daylight. Two sensors are fixed on each axis, and the output of each group is used to calculate the voltage difference. Then, the output is fed into the PLC program to move the photovoltaic panel to an optimal position. The Soft Comfort V6.1 software was used to develop the control software. A pyrometer device was used to measure the solar radiation, and a software-based online tracking method was used to determine the solar radiation. The proposed tracking system was tested and implemented for three days (May 10 to 12). The results were collected, averaged, and plotted to obtain the solar radiation curves, as shown in Fig. 18. The solar radiation collected using the two-axis tracking system was higher than the solar radiation collected using the fixed system.

Tudorache et al. [32] designed a single-axis solar tracking system that automatically searches for the optimum position with respect to the sun. This solar tracking system optimizes the conversion of solar energy into electricity by using a drive unit that receives input signals from a light intensity sensor.

The system was experimentally implemented using several hardware components, including a DC electric motor, a mini photovoltaic module, a PLC, an intelligent drive unit, and two simple light intensity sensors. The light intensity sensors were executed using two luminescent diodes of the LED type, which generate a signal that is correlated with the light intensity applied to the photovoltaic panel. The main idea is to use the light intensity sensors to receive signals from the two LEDs, generate measurements based on these signals, and find the difference to determine whether the panel should be moved to the right or to the

left. Aside from simplicity and flexibility, this system offers several advantages because the results are also optimal in terms of the cost/performance ratio. Moreover, the energy consumption is minimal because the panel is moved only in justified cases to eliminate any unnecessary consumption of energy. This system can also quickly find the position of the sun, especially at sunrise or after cloud screening, successfully works under variable light intensity conditions. By contrast, this system was implemented and experimentally tested using a mini photovoltaic module. The results slightly changed when this technique was applied in a solar array. Optical sensors can be affected by reflected or scattered light originating from surrounding obstacles. Therefore, protecting the sensors is necessary.

Deb et al. [24] used a solar tracking system to extract solar energy. The idea was to propose a single-axis solar tracking system that can be directly positioned toward the sun to optimize the conversion of solar energy into electricity. In this proposed solar tracking system, a solar tracker algorithm is utilized to determine the best angle to track the sun. Two sensors are used to measure the temperature in eastern and western locations. Then, the data are directly provided to a logic program called LabVIEW, which uses these temperatures to calculate light intensity. If the light intensity in one direction is greater than that in the other direction, then the difference between the intensities in both directions is calculated to determine the movement direction of the solar tracker system through a stepper motor directly connected to the photovoltaic module.

Pattanasethanon et al. [69] used the photo sensor system and the shading method to detect solar energy. This proposed tracking system can be utilized to direct the photodiode to the position of the sun every 10 minutes. Several electronic and mechanical devices were used to build this tracker. A phototransistor was used to detect the exact solar position by using a shade that blocks the screen of the photo sensor. The highest part of the screen can determine the sensitivity or period of tracking by this device. This tracking system has six phototransistors, and the operation of this device depends on the computation of the height of the sun shade. The proposed tracking system can only track the sun under a clear sky and good weather conditions.

4.2. Microprocessor driver systems

Microprocessors and computer systems can easily drive solar tracking systems to the position of the sun. Instead of using sensors, these systems use algorithms and mathematical equations to determine the exact position of the sun. Cheap microprocessors, such as the programmable interface controller (PIC) and the Arduino microprocessor, are used in numerous systems. The microprocessors can follow a tracking algorithm that uses the information and parameters to analyze and identify the suitable movement. Roth et al. [70] divided microcontrollers into two primary modes: clock and sun modes. The clock mode calculates the position of the sun and modifies the algorithm based on the solar error sensors, whereas the algorithm actively positions the solar panels in the sun mode. Sensors can be used in microprocessor-based tracking systems to reduce the error or calibrate the system.

Kamala et al. [71] designed a single-axis solar tracking system that aims to maximize energy harvesting. This designed tracking system was experimentally tested using two photovoltaics. The photovoltaics are driven by a PIC microcontroller based on a tracking algorithm for economic and maximum power harvesting. The photovoltaics are arranged in the form of a triangle located opposite of each other. The idea is to detect the movement of the sun based on the variation in the light intensity of the services of both photovoltaics. The power sensed by the photovoltaics can be directly measured and converted into an electronic voltage signal using a voltage divider circuit that contains two fixed resistances and a potentiometer. A differential amplifier low-side current sensing is used to sense the current of the photovoltaics. The values of voltage and current are fed to the microcontroller to calculate the output power and display this output on a liquid crystal display. Two levels of the threshold are used to drive the photovoltaics. The first threshold is utilized to activate tracking and indicate the availability of solar power. The second threshold is adopted to switch off the peripherals during the non-availability of solar power for long periods in cases with cloudy or rainy weather. The triangle form of the proposed tracking system increases its sensitivity to ambient light and its suitability for practical applications. Furthermore, this form eliminates the requirement of an external analog to digital converter. This system was also built with commercially available low-cost components; however, the problem is that this system stops working after sunset. The panel needs to be brought back to the east side to initiate tracking the next day [71].

Chabuk et al. [72], designed and implemented a microcontroller-based dual-axis solar tracking system. A stepper motor was used to rotate the photovoltaic module from one side to another, and a real-time clock microcontroller was used to find the suitable angles and feed them directly to the stepper motor. The microcontroller depends on using an algorithm that can calculate the positions and directions of the solar panel. This algorithm depends on compute the difference between both microcontroller time and the displayed time. The difference between both times will determine the next movement and direction. This system can work properly in bad weather and different locations because it does not depend on the clearness of the sky or the position of the sun. The results obtained from the system are much better than those from single axis and fixed panels. However, the algorithm used in the system cannot determine the exact position of the sun. A few cases were used to test the algorithm. The system can move horizontally with a limitation of 50°.

Ranganathan et al. [73] designed a novel digital signal processing solar tracking system. A novel algorithm was used to optimize the overall output power of the proposed solar tracking system. This novel algorithm was based on adjusting the position angles of the solar photovoltaic on the elevation and azimuth axes. The optimal angles for the solar photovoltaic were computed numerous times using the adaptive gradient ascent method until the incident solar radiation was maximized, thereby maximizing the power output as well. Taylor's series approximation was employed to generate a unique optimal position angle increment/decrement at each iteration and eliminate the need for additional sensing hardware and intensive computational resources. The proposed tracking system was built using 16-bit digital signal microcontrollers of type dsPIC33FJ32MC204 and a Microchip dsPICDEM MCSM stepper motor to apply automatic movement with Digital Signal Processing (DSP). The tracking system was simulated. The proposed technique demonstrated fast convergence and excellent tracking accuracy at all times of the day. The corresponding output power achieved by the proposed system is illustrated and compared to fixed tracking at 30°, as shown in Fig. 20. The proposed system is a novel DSP-based system. However, in addition to the substantial amount of calculation that requires time and the energy consumption for processing, several measurements must be made throughout the day.

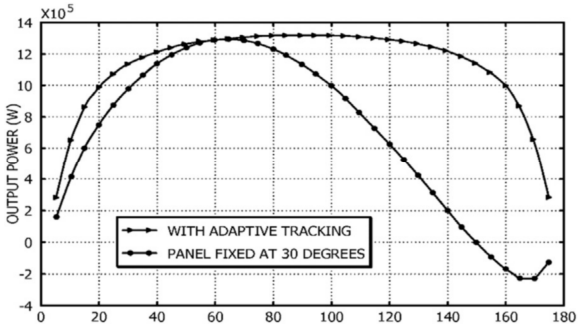


Fig. 20. Output power vs. solar elevation angle [73].

Algarín et al. [74] designed and implemented new a dual- axis solar tracking system based on real-time measurements for solar radiation. Radiation sensors have been used to send signals to high performance 16-bit digital signal controller DSPIC33FJ202MC. The obtained signals were compared with the output of mathematical algorithm to the positions of the sun, and set the photovoltaic module. A mechanical gear system was used to move the proposed system by using two motors. The two motors can move the proposed system toward the declination and hour angle. The proposed system was simulated using SimMechanics simulation tool. The system was simulated using voltage, current, and power generation, and the simulation result obtained better performance compared to fix panel. Meanwhile, a resister load of 10Ω was used to test the real-based system. The power that generated for dual-axis tracking system was much better than that of the fixed panel, and the gain was 20% for sunny days. The performance of the system in cloudy days was low. Moreover, only 10 samples were tested and evaluated, and the system was simulated without considering the effect of reactive power that resulted from the use of different electrical and electronic components. Using motors and sensors causes extra cost and more energy consumption.

Rockwell Automation [75] is one of the proposed microcontroller systems that can be utilized to control the solar tracking systems. Rockwell Automation works based on the Alten–Bradley Micrologicx controller. This controller can directly position the solar panels to the sun to capture energy. A positioning algorithm can be used to find the optimum tracker position using mathematical equations. The idea is to find the optimum zenith, vertical rotation, and azimuth angles to determine the horizontal rotation of the solar panels. Rockwell Automation can find several solutions to capture optimum solar power from the tracking system. This automation can also be used in single-axis and dual-axis trackers. The main advantage of the system is that the sun can be tracked with high accuracy to reduce the development time required to find optimum positions.

The Rockwell Automation system is divided into two parts: data collected, which are saved in a database and can be used to provide historical data; and the solar tracker system, which can directly provide the output to the positioning system.

Samantha et al. [76] designed a single-axis solar tracking system that can maximize the efficiency of photovoltaic cells by optimizing the use of solar energy. This solar tracking system is called a chronological solar tracking system because it can direct the photovoltaic panel to track the position of the sun with the assistance of a motor controlled by a microcontroller that rotates with the directions of the sun. A PIC18252 microcontroller is used by the solar photovoltaic to track the position of the sun. The rays of the sun should always perpendicularly fall on the panel because only perpendicular rays can produce maximum-intensity of solar energy. The PIC18252 microcontroller is connected to an L293D motor driver to move the photovoltaic to the desired directions. This system was developed by using a small prototype, and it can successfully track the position of the sun with a highly efficient production output.

Lakeou [28] designed a low-cost solar tracking system. This solar tracking system is connected to a 1-kW wind turbine controlled by the proposed solar tracking system. This proposed solar tracking system uses an extended cycle battery storage device, a charge controller, and an inverter. This system focuses on two essential features: Azimuth tracking to adjust the tilt angle of the surface of the photovoltaic array and tracking of the maximum solar radiation incidence, both can affect the tracking of the sun to obtain maximum energy. The success of the proposed system mainly depends on several illustrations. One of these illustrations is the positioning of the fixed solar panels based on the latitude location of the solar panels. Another important illustration involves collecting results to design a programmable controller. The designed programmable controller comprises two 24 V DC gear motors and a dual-axis photovoltaic array. The proposed system was tested and evaluated throughout an entire year. Washington City was selected as the site for testing and evaluation. The results were effective and flexible, and the hardware and software used in the system were economical.

Many studies have used the combination of photo sensors and microprocessors to drive the solar tracking systems. Rahman et al. [30] designed and implemented a dual-axis solar tracking system that can maximize the power output. This system was divided into light sensing, light comparison, control, and movement adjustment units. The light sensing unit comprises five sensors to measure the light intensity and generate a corresponding analog voltage signal. Two pairs of LDRs were used as sensors to track the exact position of the sun. One of these pairs is fixed to the vertical axis, and the other pair is fixed to the horizontal axis. The light comparison unit comprises diodes and a comparator circuit. This unit can receive the analog voltage signal from the light sensing unit. Two dual comparator ICs (LM1458) are adopted to compare the voltage levels of the horizontal and vertical axes. Diodes were used to neglect the reverse voltage drop between 0 to +5 V.

A microcontroller that can receive the output of the light comparison unit to determine the movement direction of the motors in both horizontal and vertical axes is the main unit of this system. ATmega32 microcontroller is used to organize the movement adjustment unit, which comprises two driver circuits and two geared stepper motors. The idea of the driver circuits is to turn the stepper motors in the required directions. The system remains off during the night to save energy because the motors and the microcontroller consume the saved energy. The proposed tracking system is cheap and accurate. The energy gain of this system is 52.78% compared with that of static systems. Fig. 21 shows a comparison of power collection for both static and tracking panels. The figure shows that the tracking panel gains approximately 50% more power compared with the static panel in both mornings and evenings.

Sathis [77] designed a microcontroller-based dual-axis solar tracking system. The proposed system can maximize the power incident from the sun through the photovoltaic module. The proposed system contains several components, including Arduino ATmega32s microcontroller, LDRs, stepper motors, drive motors, and Op-Amp comparator. The Op-Amp comparator is used to collect the LDRs readings and to pass them toward the microcontroller. The microcontroller can determine the suitable directions of solar photovoltaic module based on the comparator signal. The stepper and drive motors are used to move the photovoltaic module from one position to another. Fixed and tracking systems were tested to evaluate the effectiveness of using tracking systems compared with static systems. The designed system is cheap and it can increase the gained power by 40% and 45% for fixed and dual-axis tracking systems, respectively. However, the proposed system cannot work in a cloudy or dark sky, and the cost of designing such system is relatively high because of the high energy consumption, in addition to the maintenance cost.

Bharat et al. [78] proposed a dual-axis solar tracking system that can be derived by Arduino ATmega32s microcontroller. Two stepper motors are used to move the photovoltaic module toward the desired

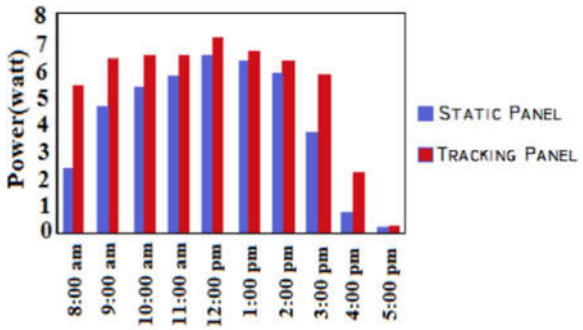


Fig. 21. Power vs. time bar diagram for both the fixed and tracking panel [30].

directions. Two pairs of LDR sensors are used to specify the location of the sun in the sky. The signals of LDRs can be utilized from the microcontroller to find the optimal directions of the photovoltaic module. The proposed model is software-based system that programmed using C++ programming language. The proposed system obtaines 45% more energy than the static systems. Although using sensors and microcontroller devices increases the cost and consumed energy and despite the fact that system can work in clear sky, the system is useful and can easily be implemented.

Akbar et al. [79] designed and implemented a dual-axis tracking system by using an AVR microcontroller. The proposed system consists of ATmega328 microcontroller, two DC motors, four light sensors LDR, and four relays. The LDRs are used to sense the light intensity of the sun and send the signal to the microcannula, which compares the obtained signals from LDRs to determine the rotating directions. Two LDRs, one DC motor and 2 relays are used to track the sun horizontally. The other components are used to track the sun vertically. Single-axis and dual-axis tracking systems were tested for a month from 9 a.m. to 4 p.m. in Kirkuk city, and the effect of both temperature and PV cover on the obtained output power was analyzed. The output power of single-axis tracking gained 24.05% more energy than the fixed panel, whereas dual-axis tracking obtained 26.22% higher energy than fixed panel. Temperature and PV cover decreased the obtained output power. However, the proposed system was tested within a very short period (one month only) Hence, other environmental conditions were not evaluated sufficiently. In addition, using LDRs with DC motors will increase the maintenance cost and the power consumption.

Otieno et al. [65] designed and implemented a dual-axis solar tracking system. A comparison between single-axis tracker systems and dual-axis tracker systems was presented to select the suitable methodology. The performance of the dual-axis method was better than that of the single-axis method, and the former method could track the sun in all directions (from east to west and north to south) with an additional gained power of up to 40%. Numerous hardware components were used to build the proposed dual-axis tracker system, including two servo motors to move the solar panel, LDR sensors to sense the sun, and the ARDUINO controller to control the servo motors. The light incident was captured by the panel and converted into electrical energy that charges a battery that can be used to save energy. A prototype tracker system was constructed using all hardware components to build a dual-axis model. A software using the ARDUINO UNO programming language was used to control the movement of the proposed prototype. The efficiency of the model was tested. The model reduced the payback time and increased the generated power by 40%.

Ghassoul [80] designed an automatic solar tracking system to maximize the gained energy. PIC was used as a control unit, and two main mechanisms were applied to implement this tracking model. The two mechanisms were a search mechanism to locate the position of the sun and a mechanism of intelligent panels, which was only used when the maximum energy is extracted.

The proposed system was designed and implemented using two LDR sensors, one of which was mounted on the solar panel, while the other was mounted on the motor. The control drive motors were used to turn and move the solar panels to the desired directions. LDRs were used to detect the sunrise and the sunset and were directly connected to Pic 18F422 microcontroller to provide the movement order to the system. A microchip assembly programming language was used to write the code for the microcontroller. All hardware components were connected together to build the prototype, and this proposed prototype was tested indoors using a mobile lamp as the source of light. The system could also perfectly align itself during both sunrise and sunset. Moreover, the system was cost effective and took days of variation through the year.

4.3. Open–closed loop driver systems

Open and closed-loop controllers can be used in tracker control algorithms. The closed-loop controller is better than the open-loop controller because it can eliminate errors that originate from the variability in insulation. A closed-loop system can track the sun using sensors. The sensors can sense the position of the sun at all times of the day. The sensed position is used in mathematical equations to make a few calculations. The results of these calculations can be used to find the next tracker direction. By contrast, an open-loop system can track the sun without using sensors. Open-loop systems use control device motors or actuators to follow the sun based on mathematical equations. These systems use algorithms based on the control of date and time without using any feedback to evaluate the results or the actions of the trackers. Moreover, the open-loop system is generally simpler and cheaper than the closed-loop system. An open-loop system does not monitor the results of the controlling process and cannot correct errors or compensate for the disruption of the system. Meanwhile, closed-loop controlling depends on driving the solar tracking system after using the sensor readings of the position of the sun or calculating the position of the sun using formulas, algorithms, and astronomical information.

Sebastijan et al. [81] designed and implemented open-loop solar tracking system that can be derived by using four light dependent sensors. The LDRs were used to determine the position and direction of the solar photovoltaic module. Two pairs of LDRs were fixed to the solar panels, and the differences between the measured current signal by LDRs in each group were utilized to trigger DC motors. The DC motors could move horizontally and vertically toward the selected azimuth and zenith angles. Fixed panels and the proposed system were tested for a few selected sunny days. The output energy of the proposed system increased by 27% compared with that of the fixed panel. The proposed system obtained 3.6% and 43% difference for the azimuth and zenith angles.

Sidek et al. [82] designed and implemented a dual-axis open loop solar tracking system that can automatically position itself by using a Global Positioning System(GPS).The proposed system used the sun trajectory path algorithm to position the solar trackers due to the sun position in the sky. Several devices were used to build the proposed system, including altitude and azimuth actuators, altitude and azimuth sensors, two DC motors, two PIC18F4431 microcontrollers, and PIC 18F4680 microcontroller. The GPS sensor was utilized to determine the exact position of the solar panel by using the longitude and latitude lines, and the digital compass sensor was adopted to determine the earth magnetic field. The PIC18F4431 microcontroller can use these measurements in addition to the day number and solar time to calculate the trajectory path of the sun. The difference between the calculated numbers and the numbers recorded in the database can be used to determine azimuth and altitude. The positions of the azimuth and altitude are utilized to directly feed the positioning controller that can move the motors either clockwise or counter clockwise. Testing and evaluating the proposed models were implemented by using Davis Weather Station Vantage Pro2(DWS) to take data The results showed that it can work properly in a cloudy sky, and can generate 26.9% and 12.8% more energy output compared with fixed panels in clear and cloudy sky conditions. The system can be easily deployed in any location, and it can be useful for mobile tracking. However, the proposed system does not work well in heavy-rain days because it depends directly on the temperature of the sky. Implementing such system is complex and expensive, and such system needs extensive calculations.

Fonseca et al. [83] proposed a dual-axis solar tracking system that uses GPS and sun positioning algorithm. The idea of this proposed system is to find the sun trajectory in the sky, azimuth, altitude, longitude, latitude, sun rise time, sun set time, and actual time. All of these variables can be determined by using an algorithm. The sun positioning algorithm depends on reading the date and time from a laptop. An ND105C GPS dongle device is used to read other variables. The algorithm can employ all the collected variables to calculate the sun angles, including longitude, latitude, sunrise, sunset, azimuth, and altitude, in addition to finding the sun trajectory in the sky. The proposed system

was tested in amny locations. It can work without used intervention, which results in high effectiveness and error avoidance. However, the system was tested for a few days only, and only the results for one day were presented. Furthermore, using simulation and calculations cannot evaluate bad weather situations.

Alexandru [84] designed and developed a novel dual-axis equatorial tracking system that can be used to maximize the degree of use of solar radiation. This system employs the daily elevation angle as an input, and the output is the torques of the motors. The mechanical model and control strategy were utilized to build a physical prototype, which was manufactured and installed in the Green Energy Independent University Campus. The proposed open-loop tracking strategy was implemented using the NI LabVIEW platform and evaluated using mathematical formulas. The error rate was measured to evaluate the robustness the tracking system. The error rate was within acceptable limits and was observed in the initial and final stages of the force action. Moreover, the stationary error was extremely low, validating the efficiency of the system.

Juang et al. [27] designed and built a closed-loop dual-axis solar tracking system to collect energy from the sun and directly convert it into electricity. The idea of this solar tracking system is to eliminate or at least control any kind of source errors in its software and hardware. This system can make decisions to increase efficiency and ensure safety. Furthermore, it can maximize irradiance and limit the battery voltage consumption to a certain value. Numerous hardware components, including two microcontrollers, four LDRs, two 24-linear actuators, an Arduino Uno R3, and a Pololu Dual Motor Shield, were used to build this proposed system The LDRs are used to provide voltage readings and compare the differences with the dead band value, thus providing a reference value for the calculated voltage difference. If the absolute value of the calculated difference is greater than the dead band value, then the linear actuators are adjusted; otherwise, a one-minute delay is applied, and then the procedure is repeated from the beginning. The motor controller extends or retracts the linear actuators until the voltage difference becomes zero. The value of the signal difference is utilized to measure the error signal, which is used to drive a power conditioner that interfaces the photovoltaic to the load. The proposed system was tested and implemented for real-time responsiveness, and the increase in power output efficiency was at least between 15% and 20%. A few solar tracking systems can be driven based on a hybrid system or a combination of open-loop and closed-loop driving methods. Rubio et al. [85] proposed a new two-axis hybrid sun tracking system that comprises a combination of open-loop tracking strategies based on solar movement models and closed-loop strategies using a dynamic feedback controller. Simulation and experimental results were presented through a comparison between a classic open-loop tracking method and the proposed hybrid system. An open-loop tracking strategy that includes the use of a microprocessor controller was employed to build the proposed model. A closed-loop strategy that includes the use of an electro-optical controller was utilized in addition to the dynamic feedback for the generated power measurements. The proposed system was designed to be an autonomous system by enabling the search mode to operate when the tracking error was excessively large. Moreover, a reduced table of errors updated every half an hour when the stored radiation was sufficient was added to prevent the system from being in the search mode for numerous times. Two modes were implemented to successfully apply this system. One of the modes normally tracks the position of the sun, and the other mode normally searches for the sun when it cannot be seen because of a few external disturbances, such as bad weather.

The proposed hybrid tracking system comprises a combination of a feed forward control strategy (open-loop) and a feedback controller strategy (closed-loop). The feedback controller can correct the tracking errors caused by the feed forward controller operating in the open loop mode. The Platform Solar de Almeria algorithm was employed to implement this proposed model because this algorithm is an improved

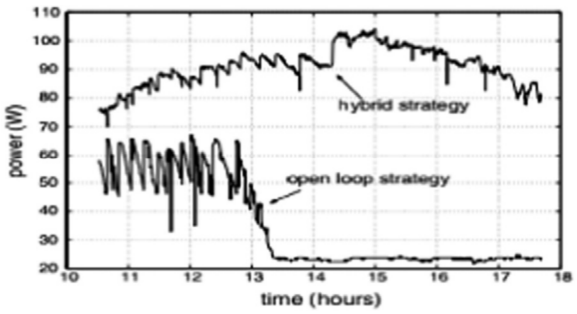


Fig. 22. Electric power generated in open-loop compared to hybrid strategies [85].

calculation of universal time for leap years and it can eliminate unnecessary operations Moreover, its calculation is quick and robust. The proposed hybrid tracking system was tested and evaluated using both simulation and experimental results. The system provided small sun-tracking errors compared with classic open-loop systems [85]. Fig. 22 shows a comparison between the electric power generated by using open-loop and closed-loop systems.

Melo et al. [86] designed and implemented two dual axis solar tracking systems based on closed- and open-loop drivers. The proposed closed-loop system depends on moving the solar photovoltaic module from one side to another based by using light sensors LDRs. An algorithm was used to compare the light intensity obtained from the LDRs and determine the new rotation of the photovoltaic module. Another algorithm was utilized in the proposed open-loop system. This algorithm works by using a fixed orientation of the sun that depends on one-year simulation for measured solar radiation under a clear sky at fixed intervals. The open- and closed-loop system were tested for different 25 days. The energy produced by closed-loop was 33% higher than that of fixed panel and 28% higher than that for the open loop system.

4.4. Intelligent driver systems

Systems Artificial intelligence (AI) has been used to drive the solar tracking systems [87–93].Several AI techniques, such as fuzzy logic, neural networks, and a combination of both called fuzzy neural networks, have been utilized to control solar tracking system models [48,94–100]. A solar tracking system based on a connection between the photovoltaic and a hybrid mix of the electric load also was designed [101,102]. This system is controlled by fuzzy logic, which can detect and track the power point to find the Maximum Power Point (MPP) by using the temperature change of the load [103]. A mathematical framework MATLAB/Simulink software environment was used to simulate the photovoltaic panel, which was connected to a DC load. Two control units were used to detect the MPP using both the voltage control of the DC load and the speed of the permanent magnet DC motor using DC/DC choppers. A few errors were found when the different points of MPP operations were tested.

Alata et al. [104] developed a solar tracking system using a fuzzy control. In addition to the fuzzy if–then rules to model the inputs and outputs of the system, a Sugeno fuzzy inference system was utilized to model the control system. A 3D virtual reality simulator was employed to simulate this system because it allows the system to perform as it would in actual conditions. Real values were adopted as simulation parameters. This tracking system was used every four minutes to track the 1° movement of the sun.

Iqdour et al. [35] proposed a solar tracking system that can predict the daily solar radiation in Marrakech, Morocco. A Sugeno fuzzy inference system was employed to model daily radiation. The system and its results were compared with two other models based on high-order statistics. The proposed fuzzy logic model demonstrated better than the others in predicting daily solar radiation.

Takun et al. [105] used a fuzzy logic controller to find the MPP

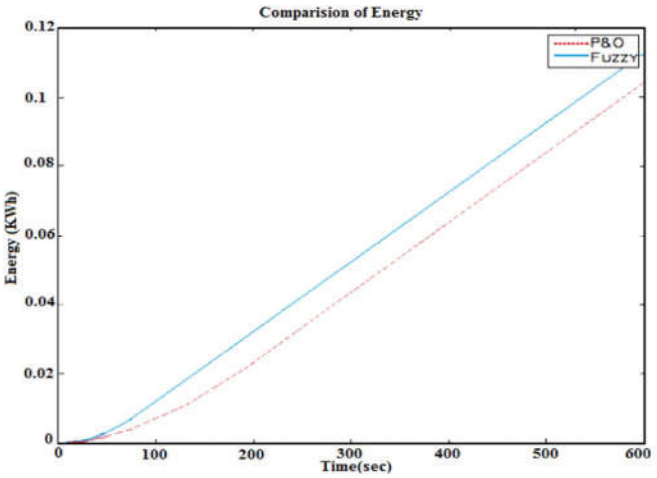


Fig. 23. Energy obtained from FLC and PO [105].

using a photovoltaic system. This proposed controller was compared with the conventional Perturb and Observation technique (PO). The performance obtained using fuzzy logic control was better than using the PO method. Furthermore, the system itself was simple. The proposed system was divided into three stages: fuzzification, rule-based, and defuzzification. The measured changes in power and current were adopted as inputs for the fuzzy logic control system, and the output was the value of the referenced current, which is the command for controlling the current obtained from the photovoltaic. The proposed system was modeled and simulated using a mathematical framework MATLAB/Simulink under a constant temperature and with varying isolation. The simulation results showed that this system is better, faster, and more accurate compared with the PO method. Fig. 23 shows a comparison of energy between the fuzzy logic controller (FLC) and the PO controllers. However, the work has not been practically implemented or tested until now, and numerous missing parameters were not considered during system design. Thus, this system is less efficient and less reliable than other proposed systems.

Sendoya et al. [106] designed and implemented a dual-axis solar tracking system that can be controlled by fuzzy logic. The idea of the proposed model is to find the prependicular position of the sun on the solar panels. The position of the sun is determined by using four LDRs that can measure light intensity, voltage, current, and temprature. Two LDRs measure the light intensity horizontally, and two measure the light intensity vertically. The difference between the measured light intensity is used to determine the position and direction of the solar panels. Three solar photovolatic modules and Yeasu G5500 roto motor were utilized to build the proposed system, and fuzzy logic control was used to drive the solar photovoltaic modules. The Mamadani fuzzy inference system was employed to position and rotate the photovoltaic modules, and feed the rotor of Yeasu G5500. The rules of the fuzzy control were selected based on the differences between the horizontal and vertical light intensity. System testing covered the path of the sun during 12 hours, and the output power of the proposed model increased by 12.45% compared with that of fixed solar panels.However, the proposed system is very expensive because three solar panels of 300 w and the expensive rotor device Yaesu G5500 are used.

Xinhong et al. [107] used fuzzy logic to control the solar tracking system. The rules of this tracking system were obtained from a real experiment. A few algorithms were utilized to create a self-learning system and build the system rules. The structure of the fuzzy logic employed in this system comprised three stages: fuzzification, rule based, and defuzzification. The system was practically built, tested, and simulated using a few software program in addition to a few indoor light sources. However, this system was tested indoors by using a lamp to act as the sun.

Zaher et al. [108] proposed and designed a dual axis solar tracking system using fuzzy logic controller and METEOSAT image processing. The proposed model depends on the use of image processing to detect the cloud coverage in the sky and cloud duration. Cloud coverage and duration were utilized as inputs for the fuzzy logic controller, and the output was the optimal position of the solar panel. Three different cases, including the clear cloudy, and partially cloudy sky, were examined. The proposed fuzzy logic controller was utilized to implement a partially cloud sky. It can decide on the suitable position that can gain high efficiency. Four images were obtained from satellite every hour, and an algorithm based on image processing was used to compare the captured images and find the optimum position. The decision of the algorithm can be sent directly to a microcontroller by using RS232 serial communication. The microcontroller was used to decide on the optimum movements based on fuzzy logic. The proposed system was tested for 17 months, and the output showed that the proposed system can work well in partly cloudy skies and it can gain 23% increased output.

Huang et al. [26] designed and implemented a solar tracking generating power system. An expert controller, sensors, and input/output inference were used to build this system. A fuzzy algorithm was developed to build the controller of the proposed system. The proposed solar tracking system was connected to a field-programmable gate array (FPGA) control system. Light sensitive sensors were used to provide a signal to the FPGA through an analog/digital converter. Two driver motors were employed to build the solar tracking system. These two motors worked independently to minimize the power consumption of the system. The fuzzy controller was divided into four phases: fuzzification, control rule based, fuzzy inference, and defuzzification. The idea is to use the signal that goes through the analog/digital converter when a difference exists between the eastward–westward direction voltage and the southward–northward direction voltage. The FLC uses these differences to produce pulses to motor drivers, and the motor drivers produce a signal to move the step motor to the desired angles.

This model was designed and implemented for four solar cells panels. Every two solar panels were connected in a series to create two different sets connected in parallel as a solar array. The problem in this connection is the numerous differences in the characteristics of solar panels. Moreover, the changes in environmental factors, such as dust and clouds, can affect the results. These changes also affect the values of the measured currents and voltages in the different sets. Obtaining a high temperature for one set compared with the other is another problem because when these sets are connected in parallel, one of them acts as a load for the other. The experiment for testing this proposed model was implemented on the roof of the third building in Yuan Ze University, and two batteries of 12 Volts and 7 Ah were used to collect the generated power. Two methods were tested: fixed tracking and solar tracking. The comparison showed that the efficiency of the solar tracking system is 6.7% higher than the efficiency of the fixed tracking method (Fig. 24).

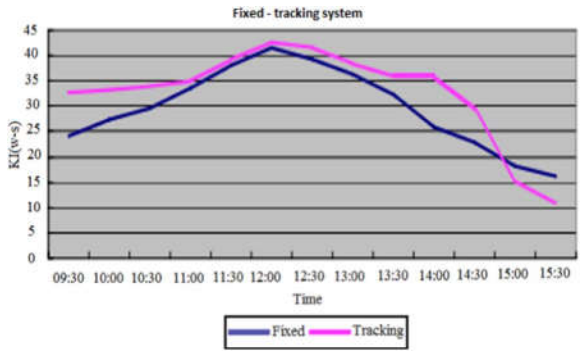


Fig. 24. Power generation comparison of fixed angle and FPGA [26].

Aldair et al. [109] developed a dual-axis solar tracking system based on using a neuro-fuzzy controller. The FPGA controller was designed with two optical photo sensors to orient the photovoltaic module, two linear actuators to move the system, and a drive motor. The proposed system was utilized to compute the difference between the errors measured by the photo sensors. The difference and summation of errors were employed as inputs for the proposed fuzzy neuro controller, and the output was the directions fed directly to the drive motor in reference to the threshold values. Three days on January were used to collect the data set for testing the proposed model. The system was evaluated using MATLAB Simulink and tested practically. The proposed controller was more robust than the fuzzy logic controller. It could track the sun trajectory in the sky. The Dual-axis system achieved 50.6% more energy than fixed panel. However, using one or two hours to test the model in sunny or partly cloudy day cannot guarantee that the system is works well. Moreover, switching the driving motor 11 times in one hour consumes the collected energy and can affect the results of the system.

Chen et al. [110] proposed a dual-axis solar tracking system using a FPGA controller to maximize the generated power. The proposed FPGA controller was built by taking measurements from four different sensors. The FPGA controller was utilized to find the suitable direction of the photovoltaic module by computing the error difference between each pair of sensors (east-west and north-south). A threshold value of 0.1 was adopted to determine the suitable direction of the photovoltaic module, and a servo motor moved the system to the desired directions. MATLAB-Simulink was used to evaluate the performance of the proposed system. The system was tested experimentally indoors by using a Lead-Acid battery to generate power for the tracking system instead of using the sun radiation influenced by illumination. Thus, the proposed system was unstable. The proposed system can be installed in various locations, and it sets off at night to save the collected power. However, the measurements and processes of the system were repeated every 10 minutes and thus caused high energy consumption.

Chen et al. [111] developed a dual-axis solar tracking system based on self-sufficient solar power generation and the FPGA system to improve the temperature rise and increase the gained power. The proposed FPGA was used for data acquisition, and the LABVIEW program was employed for data feedback, calculation, and display. The data of temperature in different charging modes were collected experimentally every 2 seconds. The proposed system improved the temperature rise, and increased the efficiency of charging compared with traditional methods. The proposed system can avoid the rapid aging of the battery, and it can be self-sufficient.

AL-Rousan et al. [112] proposed another fuzzy logic model to track the sun by predicting the suitable angles that provide maximum outputs (current, voltage, and power) at a specific time on different days. Mamdani and Sugeno inferences were tested to model this system. The model could predict both the tilt and orientation angles that successfully measured the maximum current, voltage, and power. This model was developed using a mathematical framework MATLAB tool, thereby making it a hardware-software-based model. The code to program this model was written with the C programming language. The proposed fuzzy logic-based model comprises three inputs and two outputs. Time, day, and month are the inputs of this model, and the outputs are tilt and orientation angles. Fuzzy sets and their membership functions were defined to belong to the inputs and outputs, and triangular and trapezoidal shapes were used to draw the sets of membership for each variable. Fuzzy rules were were utilized to characterize the fuzzy system and connect between the inputs to correctly provide the output. Different fuzzy rules were found using the data collected from a long experiment to manually test the real tracker system. Three techniques were adopted to collect data: fixing the tracker system to the south, tracking the position of the sun, and the no side shadow. Sugeno and Mamdani fuzzy inferences were proposed to model and test this system [113], and the model was used to find the power output. The model was evaluated to determine the correct values compared with the experimental results. The error ratio of the tested tilt angles was approximately 0.051% on the average and approximately 0.063% for the tested orientation angles. These results show that the proposed model is effective and can be used in the Jordan region or nearby countries.

Huang et al. [114] designed and implemented a dual-axis solar tracking system using a fuzzy logic controller. Several devices, including lead acid batteries, gear box, stepper motor, and light detection circuit, were utilized to build the proposed model. A fuzzy logic controller was employed to determine the suitable time to track the sun. Sunlight voltage and solar cell current were used as inputs to the fuzzy logic system, and the output was the motor motion. The fuzzy controller employed the produced current in the solar panel to determine the suitable time to move the solar panel by the motor. A threshold of 0.5 for the fuzzy output was used to determine whether the motor will work or not, and a database for the angle values was utilized to determine the next angle. To move the panel, a LabVIEW program was used to calculate the suitable pulse that can feed the motor to move toward the desired angle. A MATLAB simulation was performed to build the fuzzy controller. Moreover, another solar panel derived by sensors was adopted to compare the two technologies. The proposed fuzzy logic controller performed better than the sensor-based system; its capacity and output power increased by 36% and 4.32 watt respectively. The proposed system worked properly in bad weather and reduced the number of times the motor was used to rotate the solar panels. Thus, unnecessary energy consumption was reduced.

The neural network technique is another technique to control a solar tracking system [115,116]. Cheikh et al. [117] proposed a control method to increase the output power gained by a solar photovoltaic module. The photovoltaic generator and load were used to monitor MPP using three different variables, including solar insulation, temperature of the junction, and dynamic charging voltage. The idea of this research is to track different measurements of the current and voltage of the solar photovoltaic module to evaluate the output power of the photovoltaic module and then detect MPP. A battery is used, instead of the sun, as a power source. A few variables are also used, including the temperature of the junction, solar insulation, battery, and desired cyclic duty ratio. All of these variables correspond to train data and insert the result in a control architecture. This architecture was tested to determine the performance for MPP in a random condition. The idea is to simulate the proposed neural network method to position the photovoltaic modules. Positioning the photovoltaic modules will obtain the MPP position. The proposed model was simulated, and the obtained results were compared with those of the MPP tracking model. The results showed that this model is particularly effective in a complex and nonlinear system.

Makhloufi et al. [45] designed and implemented a tracking power photovoltaic system using an artificial neural network control strategy to maximize the gained power. The idea is to perform a simulation study of the MPP tracking method using an artificial neural network to control the photovoltaic modules. Two main control methods, MPP and PO, were discussed. A photovoltaic array model was designed using a single-diode mathematical model to act as silicon photovoltaic cells. Photocurrent source, nonlinear diode, and internal resistances were used as well. The Simulink tool in MATLAB was used to simulate the photovoltaic array system, and two different inputs, temperature and solar irradiation, were adopted to determine the output characteristics of the photovoltaic array. A protection device, namely, a boost converter, was added to obtain the voltage and current feedback from the control circuit. In this system, solar irradiation can control the DC–DC converter using the neural network method. The PO technique can use the data collected in the neural network method. A MATLAB–Simulink was also utilized to design the proposed neural network model, and real conditions were implemented to test the model after applying a variation in irradiation to determine the robustness of the proposed artificial neural network technique. The artificial neural network and PO

methods were tested and evaluated to determine their differences. Using the artificial neural network controller improved the efficiency of the overall system and minimized the energy loss. The MPP tracking artificial neural network method obtained a relatively good transient performance, it improved the response of the photovoltaic system, reduced the time response, maximized the power point, and eliminated the fluctuations around this point. However, implementing this model using a simulation does not provide real outputs.

Panait et al. [118] designed and implemented a simple neural network solar tracker system to optimize conversion efficiency in off-grid solar generators. A new generation of neural network solar trackers using the neural network and nervous chains was proposed to reduce the cost and the complexity. The light-intensity-sensing method was utilized to boost the efficiency of solar conversion by up to 100% during summer and up to 40% during winter. A tracker system that contains a stepper motor, an H-bridge, and a buffer amplifier, was built for research. The proposed neural network model was tested in a un real-time environment for evaluation.

Rabee at al. [119] developed new model that can predict the daily average solar radiation. The artificial neural network principle was used to predict solar radiation by employing the actual average solar radiation from five locations. Multi-layer feed forward neural networks were used to implement the proposed model. Data recorded for three years were utilized for the training stage, and data recorded in another year were used for the testing stage. Three models, namely, Gradient Descent error optimization ANN, Levenberg Marquart error optimization algorithm, and ANN with a large number of neurons were tested to determine the optimum solar radiation. The Mean Absolute Percentage Error (MAPE) for the proposed models were 86.3 and 85.6 for Gradient Descent ANN and Levenberg Marquart lagorithm respectively. The last model that used large number of neurons achieved 94.75% of efficiency when 1460 neurons in the hidden layer were used.

Kayri et al. [120] proposed and implemented a single axis solar tracking system using an intelligent neural network controller. The neural network model was proposed to increase the accuracy of the predicted values. Several variables, including wind speed, wind direction, solar elevation angle, air temperature, relative humidity, and global radiation, were used as input variables, and output power was used as the output variable. The feed forward back propagation neural network algorithm was utilized in addition to two error optimization methods, namely, Levenberg Marquardt and Bayesian Regularization. The proposed system was tested on cloudy and sunny days separately, and measurements for three different days wereused to build the proposed system. The proposed system produced perfect results during testing.

Thiaw et al. [121] discussed the idea of using the neural network as a controller for MPP tracking of a photovoltaic generator and to describe numerous applications by using the artificial neural network in renewable energy. The MPP and neural network tracking principles were discussed and compared. Determining the exact relation between the inputs and outputs, such as neural network principles, was unnecessary. A simulation of an efficient neural network controller was implemented to determine the optimal values of energy. An MPP tracking system using an artificial neural network technique was designed to extract and utilize the maximum portion of the incoming solar radiation [122]. The MPP method is one of the most popular methods in the world. It is the most effective method in cold weather and when the battery is low because the photovoltaic modules can still work properly. The classic PO method, which can be used to observe the output results during the changing of the directions of photovoltaic modules, was discussed as well. Other algorithms, such as an incremental conductance algorithm (which can perform MPP under the varying irradiation and daily conditions) and a constant voltage algorithm (which sets the current of the photovoltaic modules to zero to measure the open circuit voltage), were also examined.

A new algorithm was proposed to improve the tracking speed of the PO method by using current–voltage curves and their MPP. A simulation in MATLAB was performed to build the photovoltaic module in addition to the proposed model (Buck-Boost) converter, which is a type of DC-DC converter that has an output voltage greater than or less than the magnitude of the input voltage.

Yang et al. [123] developed artificial neural network vector control system. This system was proposed to evaluate the performance of using a neural network in solar photovoltaic systems. Several components, including a DC-DC convertor were used to design the proposed system. The proposed feed forward neural network controller depends on four different inputs and two hidden layers with six neurons in each layer. The proposed system was evaluated through a simulation in MATLAB Simulink and with hardware in the laberotary. The output of the proposed system is the ratio of the invertor output voltage generated from the DC-DC convertor. The proposed artificial neural network controller improved the performance of the photovltic module systems. It was reliable, efficient, gained more output power, and achieved a stable dc voltage, although it did not cover all the current transient conditions.

Shenawy et al. [124] investigated the logic controllers, conventional integral, and derivative controller performance to compare the artificial intelligent controllers of solar tracking systems. The idea of using a solar tracking system was explained together with the differences of using single- and dual-axis solar tracking system. Furthermore, the idea of using intelligent controllers to control these solar tracking systems, especially AI science techniques such as neural networks and FLCs, was discussed. Both techniques were simulated using Simulink in MATLAB in addition to the real-time workshop tools to understand the idea of using them as controllers. Then, these techniques were tested and evaluated to ensure the best performance compared with the other methods.

Armendariz et al. [125] presented an intelligent sun-tracking controller based on fuzzy logic to maximize the amount of energy acquired by the photovoltaic module. The idea is to achieve the maximum power of energy when maintaining the sunlight incidence direction perpendicular to the panel surface and design a fuzzy controller system for the solar panel. A dual-axis solar tracking controller was created, and Fuzzy Rules Emulated Network (FREN) controllers were used for implementation. FREN operates under principles similar to the fuzzy neural network technique. FREN was developed and applied to control various nonlinear plants. The idea is to use fuzzy logic rules by using neural network layers. In this research, four different layers were used, and the input was used to drive the output to receive the corresponding time and years. An experimental prototype was developed in the laboratory, and the controller was programmed in MATLAB.

Arulmurugan et al. [126] developed a new coupled optimized Hopefield fuzzy logic controller by using artificial neural networks. The idea of the proposed system is to maximize the power obtained from the system, generate constant voltage to the battery, and mitigate the limitation of covering partial shading conditions. This system is based on the Mamadani fuzzy logic inference system and fuzzy rules and uses the neural network system. Current and voltage were utilized as fuzzy logic controller inputs, and the reference power was used as the output. The reference current obtained by fuzzy logic was used as a neural network input to determine the photovoltaic module current and voltage. The outputs of the neural network were fed directly to DC-DC buck boost convertor. Simulation using MATLAB Simulink was applied to ensure the efficiency of the system, and an experimental test in the laboratory was implemented. The system effectively generated more power compared to other methods.

Otieno et al. [127] presented and designed a high-performance tracking system based on the maximum power delivered from a photovoltaic module by using adaptive neural fuzzy inference systems (ANFISs). ANFISs combine the fuzzy inference system and neural network layers. These systems combine the learning abilities of artificial neural networks and the capability of fuzzy logic to handle imprecise data. ANFISs can also handle time-varying and nonlinear problems. The

Table 2
Comparison between different active solar tracking system.

Technology	Characteristics	Advantages	Disadvantages
Electro-Optical Sensors [68]	• Sensor Driver System • Dual-axis system • Hardware Implemented • Used two motors	• Cheap • Rotation in both azimuth and elevation	• Applied on a prototype only
Light-Dependent Resistor [23]	• Sensor Driver System • Dual-axis system • Hardware and Software Implemented • Compared to fixed tracking • Used servo motors	• Can move • Increased the output power by 30%–40%	• Implemented only in three days 10th, 11th, and 12th of May
Light Intensity Sensors [71]	• Sensor Driver System • Single-axis system • Hardware Implemented • Used DC motors	• Cheap • Eliminated any unnecessary energy consumption	• Used mini PV module to implement the system
Temperature Sensors [24]	• Sensor Driver System • Single-axis system • Simulation Implemented using LabVIEW software • Used a stepper motor	• Cheap • Moved east to west and clockwise and counterclockwise	• No practical data
Photo Sensors [69]	• Sensor Driver System • Hardware Implemented • Compared to other models • Used a controller motor	• Moved every 1ten minutes	• Inefficient • Have a maximum error rate in the morning and evening
Digital Signal Processing System [73]	• Sensor Driver System • Single-axis system • Simulation implemented using Matlab • Compared to fixed tracking • Used a motor	• Moved to positive and negative directions • Increased the output power in Watts • Novel system	• No practical data • Measurements are necessary through the entire day • Numerous calculations are necessary
Digital signal processing [74]	• Sensor Driver System • Used 2 motors • Dual-axis solar tracker • Hardware and simulation using SimMechanics	• Increased the generated power by 20% compared to fix panel.	• 10 samples only used • Not considering the effective of reactive power appeared because of using electronic components together. • Using motors and sensors will cause extra cost and more energy consumption.
PIC microcontroller [71]	• Microprocessor Drive System • Single-axis system • Hardware Implemented • Compared with system and losses • Used a DC motor	• Cheap • Followed the position of the sun • Is better by 28.8%	• Unavailable in bad weather and after sunset • Electronic Circuit Power Consumption = 1 Watt • Motor and Driver Consumption = 10 Watt
Clock- Microcontroller [72]	• Microprocessor Drive System • Dual-axis system • Use Stepper motor • Compared to single-axis tracking • Hardware-based and simulation	• Cheap • Gain better energy • Save energy at night • Can be used for bad weather	• Not depend on the sun position • Few cases were used to test the model • Limited vertical angles used cannot exceed 50°
Rockwell Automation [75]	• Microprocessor Drive System • Both single- and dual- axis system • Hardware Implemented • Used motors	• Cheap • Moved the panel to face the sun • Tracked the sun in high accuracy to reduce the development time to find the optimal position	• Requires a database
Chronological Solar Tracking [76]	• Microprocessor Drive System • Single-axis system • Hardware Implemented • Compared to fixed tracking • Used motor	• Moved PV to the position of the sun • Increased the electric output by 30%	• Implemented only on a small prototype
Programmable Controller System [28]	• Microprocessor Drive System • Dual-axis tracking system • Hardware implemented • Compared to available models in the market • Used a DC-Gear Motor	• Cheap • Moved in two directions • Save $1000	• Requires a long data collection
LDR + ATmega 32 Microcontroller [116]	• Combination Sensor-Based Microprocessor Driver Systems • Dual-axis tracking system • Hardware implemented • Compared to static systems • Used a geared stepper motor	• Cheap • Moving • Remained off during the night • Increased the gain by 52.78%	• Implemented on a prototype only • Motors and microcontrollers consume energy
LDR + Arduino [26]	• Combination Sensor-Based Microprocessor Driver Systems • Dual-axis tracking system • Hardware implemented • Compared to single axis tracking • Used servo motor	• Minimized the power consumption operation of the system • Moved east-west and north-south • Increased the gain by 40%	• Implemented only on a prototype
LDR + PIC microcontroller [80]		• Cheap	• Implemented on a prototype only

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Table 2 (*continued*)

Technology	Characteristics	Advantages	Disadvantages
	• Combination Sensor-Based Microprocessor Driver Systems • Dual-axis tracking system • Hardware implemented • Used a motor	• The energy consumed by the panel driving motor is less than that extracted • Moved to three positions	• Tested indoor
Arduino ATmega32s + LDRs [77]	• Combination Sensor-Based Microprocessor Driver Systems • Dual-axis tracking system • Used drive and stepper motors • Software implemented	• Cheap • The gained power increased 40% and 45% for static and dual axis tracking. • Low waste and pollution	• Not working in dark and cloudy sky • Consume more energy • High maintenance cost • Implemented using software only
Arduino ATmega32s+ LDRs [78]	• Combination Sensor-Based Microprocessor Driver Systems • Dual-axis solar tracking system • Used 2 stepper motors	• Cheap • Increased the gained power by 45%	• Working only in clear sky • Cause more energy consumption • High cost
AVR microcontroller+ LDRs [79]	• Combination Sensor-Based Microprocessor Driver Systems • Used 2 DC motors • Both singl and dual axis tracking • Hardware implemented	• Low cost • Increase the generated output energy	• The test done for one month only • Using LDRs and DC motors will have increased the consumed energy. • Need clear sky to work sufficiently.
Open Loop [81]	• Open-Closed Loop Driver System • Dual-axis tracking system • Used DC motors • Hardware Implemented • Compared to fixed tracking	• Can work in bad weather • Increased the output energy by 27%	• Tested in few sunny days • The differences between calculated and measured azimuth and zenith is 3.6% and 43%
Open Loop [82]	• Open-Closed Loop Driver System • Dual-axis tracking system • Used DC motors • Hardware Implemented • Compared to fixed tracking in clear sky and heavy rain	• Can work in cloudy sky • Generate more energy compared to fixed panel • Low energy consumption • Can be deployed in any location • Suitable for mobile tracking	• Not working will in heavy ran days • Complex and expensive • Needs long processes and calculations. • Used Davis Weather Station Vantage Pro2(DWS) to take data
Open Loop [83]	• Open-Closed Loop Driver System • Used GPS system • Dual-axis tracking system • Simulation study	• Can be deployed in any location • Work without the intervention of the user (more effective and it might avoid errors)	• Tested in few days only and showed the results of one day • Cannot evaluate bad weather situations • Depends on the difference between calculations and GPS (not real)
Open Loop [84]	• Open-Closed Loop Driver System • Dual-axis tracking system • Simulated using Simulink-Matlab + CAD software Solid Work • Hardware Implemented • Compared to fixed tracking • Used a DC motor	• Cheap • Movable • Energy efficiency of 605 WH/Day • Increased the gain by 49.2%	• Implemented on a prototype • Error rate observed in the initial and final stages
Closed Loop [27]	• Open-Closed Loop Driver System • Dual-axis tracking system • Hardware implemented • Compared to other models • Used a motor	• Cheap • Movable • Used the threshold value for battery voltage consumption • Increased the power output efficiency by 15–20% • Error rate eliminated or controlled	• Totally depends on the readings of LDRs.
Hybrid [85]	• Open-Closed Loop Driver System • Dual-axis tracking system • Hardware and software implemented • Compared to classical open-loop tracking • Used control motors	• Cheap • Moved to the position of the sun • No energies consumed by moving the PV modules • Error rate is relatively small • Increased the electric power by 55%	
Closed Loop + Open Loop [86]	• Open-Closed Loop Driver System • Dual-axis tracking system • Hardware and software implemented	• The closed loop gains more energy 33% • The open loop gains 28%	• The proposed systems tested in 25 days only • The test only for clear sky
MPPD based on Fuzzy Logic [101]	• Intelligent Driver System • Dual-axis tracking system • Simulated using Matlab-Simulink • Compared to dual fuzzy logic • Compared to MPP • Used motors	• Cheap • Movable • Low error rate	• Used only the change in temperature of the load
Sugeno Fuzzy Inference [104]	• Intelligent Driver System • Dual-axis tracking system • Simulated using a 3D Virtual Reality Simulator (Simulink-Matlab) • Compared to different insulations • Used a DC motor	• Moved 1° every four minutes • Low energy consumption • Insulation in $W/m^2 = 0.002$	• High error rate in different angles
Takagi-Sugeno Fuzzy Inference [35]	• Intelligent Driver System • A simulation study	• Time was 0.12 compared to 6.34 and 5.84	• Normalized root-mean- square error is 13.97%

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Table 2 (*continued*)

Technology	Characteristics	Advantages	Disadvantages
	• Compared to a closed-form and cum4-LS		• Normalized mean bias error is 0.16%
Fuzzy Logic Controller [105]	• Intelligent Driver System • Simulated in Matlab-Simulink • Compared to conventional PO technique	• Better performance and gained power	• Simulation only applied to the driver circuit and not to the system
Fuzzy Logic Controller[106]	• Intelligent Driver System • Dual-axis tracker • Used Yaesu G5500 rotor • Implemented practically	• Gained much output power compared to fixed panel.	• Very expensive because of using Yeasu G5500 and 3 solar panels. • The electromagnetic devices will consume more energy.
FPGA [107]	• Intelligent Driver System • Dual-axis tracking system • Hardware implemented • Compared to fixed solar current collection system • Used motors	• Tracked the movement of the sun • Total average power every 24 hours is 18.1%	• Implemented only indoor using a source of light
Fuzzy Controller [108]	• Intelligent Driver System • Dual-axis tracking system • Used METEOSAT image processing • Hardware implemented • Compared to clear and overcast sky	• Increased the gained power by 23%	• Working well in partly cloudy sky only • Depends on comparing images in cloudy and clear sky (not real) • Taking 4 images every hour • Taking satellite images
FPGA [43]	• Intelligent Driver System • Dual-axis tracking system • Hardware implemented • Compared to fixed tracking • Used motors	• Track the movement of the sun • Energy consumption can be minimized by moving one motor each time • Increased the efficiency by 6.7%	• Implemented only on a small PV module
FPGA [109]	• Intelligent Driver System • Dual-axis tracking system • Hardware and software implemented • Compared to fixed tracking • Used drive motors • Used photo sensors	• The controller more robust than fuzzy logic controller • Can track the sun trajectory • Low cost system • Gain more power	• Measurements have been collected during short period time. • Data set has been collected for 3 days only. • The motor switch 11 times in one hour
FPGA [110]	• Intelligent Driver System • Dual-axis tracking system • Hardware and software implemented • Used drive motors • Used photo sensors	• Can track the sun trajectory • Set off at night to save energy • Can be installed in various locations.	• Track the sun every 10 minutes that will consume more energy. • Repeated the measurements 6 times in one hour • The system was unstable by using sun radiation • Tested indoor using battery as a source.
FPGA [111]	• Intelligent Driver System • Dual-axis tracking system • Hardware and software implemented • Used drive motors	• effective in the suppression of battery temperature rise • avoid the rapid aging of the battery • self-sufficient	• tested indoor • not taking weather conditions • repeat the measurements every 2 seconds that will consume energy.
Mamdani + Sugeno [112]	• Intelligent Driver System • Dual-axis system • Hardware implemented and simulation in Matlab • Compared to fixed tracking due south and no side shadow	• Moved horizontally and vertically • Low energy consumption • More efficient • Low error rate of the tilt angle is 0.051% and the orientation angle is 0.036%	• Long experiments and data sets are necessary
Fuzzy Logic Controller[114]	• Intelligent Driver System • Dual-axis system • Hardware implemented and simulation in MATLAB • Compared to sensor-based drive systems	• Moved horizontally and vertically • Increased the capacity, and power by 36% and 4.32 w compared to sensor based • Reduced the number of time to move the motors • Working well in bad weather	• Used threshold value to decide about moving the panel (0.5) • Depends on database to determine the suitable angles
Artificial Neural Networks [117]	• Intelligent Driver System • Simulation implemented • Compared to MMP and incremental conductance	• High gained power • Low error rate of 0.063%	• No practical data
Artificial Neural Networks [45]	• Intelligent Driver System • A simulation study • Compared to PO method	• Movable • Can minimize the energy losses • Efficient	•Simulation Study
Simple Neural Networks [118]	• Intelligent Driver System • Dual-axis tracking system • Hardware and Simulation using LabVIEW implemented • Compared to MPP	• Cheap • Movable • Efficiency of up to 100% in summer and 40% in winter	• Hardware and Simulation-LabVIEW
Neural Networks [119]	• Intelligent Driver System • Predict the daily solar radiation • Software implemented	• Achieve MAPE 94.75%	• Use solar radiation data to predict the solar radiation • Used the number of neurons to 1460 to achieve 94.75% • Need long process time
Neural Networks [120]	• Intelligent Driver System • Single-axis tracking system	• Used error optimization methods • The system shown that it can work properly	• Used data for three days only in cloudy weather.

(*continued on next page*)

Table 2 (*continued*)

Technology	Characteristics	Advantages	Disadvantages
Neural Networks [121]	• Hardware implemented • Intelligent Driver System • Dual-axis tracking system • Simulation implemented • Compared to classic MPP	• Low error • Efficient	• Used data for three days only in sunny weather. • Moved to the right direction only
Artificial Neural Networks [122]	• Intelligent Driver System • Simulated Using Matlab-Simulink • Compared to classic MPPT and classic PO	• Moved to the MPP	• No practical data
Artificial Neural Networks [123]	• Intelligent Driver System • Simulated using MATLAB-Simulink and hardware implemented • Using DC-Dc convertor	• Efficient controller to be used for maximizing the power • more reliable and efficient extraction of solar power • more stable DC-link • voltage	• Tested in laboratory • Not cover all the current transient conditions
Dynamic Fuzzy Logic Controller [124]	• Intelligent Driver System • Single-axis tracking system • Simulated using Matlab-Simulink • Compared to classic fuzzy logic, neural network MPP	• Cheap • Movable • Obtained better results	• No practical data
Fuzzy rules system astronomical yearbook [125]	• Intelligent Driver System • Dual-axis tracking system • Hardware implemented • Compared with the angles obtained from astronomical records	• Cheap • Moved the panel to the optimum directions • Obtained better results	• Practical hardware of two degrees of freedom (mini solar tracker)
Fuzzy- Neural [126]	• Intelligent Driver System • Hardware and software implemented • Compared to fuzzy logic controller • Used DC-DC buck boost converter	• The system is effective to obtain more energy • Can mitigate the disability to cover the partial shading conditions limitation	• Based on laboratory tests • Relatively complex • Need several processes and calculations
ANFIS [127]	• Intelligent Driver System • Simulated using Matlab-Simulink • Compared to fuzzy logic based MPP	• Cheap • Reduced the electricity cost • Obtained better results but closed	• Simulation using Matlab-Simulink
ANFIS [128]	• Intelligent Driver System • Software implemented • Used buck-boost convertor • Compared to FLC	• faster, simpler, and more stable compared to FLC	• Not covering all the weather conditions • Several parameters have been omitted to simplify the proposed system

model was tested and evaluated. The model could accurately track MPP and reduce the cost of consumed electricity. The performance of ANFISs was found to be higher than that of the fuzzy logic-based MPP.

Sharma et al. [128] proposed a high performance tracking system to maximize the photovoltaic module generated power by using ANFIS controller. The ANFIS controller was utilized to determine the optimum duty cycle that can be fed directly to the buck-boost convertor by using temperature and solar irradiance. Data of photovoltaic temperature and solar irradiance were collected from photovoltaic module every one second. A total of 170 readings and 49 rules were utilized to build the system. The proposed ANFIS model was evaluated with MATLAB program. The performance of the proposed system was compared with that of the fuzzy logic controller. The ANFIS model reacted faster, simpler, and more stable. However, evaluation the proposed model was implemented without covering all the weather conditions, and several parameters were omitted to simplify the proposed system which affected the performance results.

The driving method is important in controlling solar tracking systems. Selecting a suitable tracking principle and an efficient driving method to control solar tracking systems is difficult but important. Solar tracking systems can be divided into two main types, active and passive. Active solar tracking systems can be classified into numerous types. The driving method used to control the movement of solar tracking systems is one of the basis of classification. A comparison of different driving methods is presented in Table 2. The table also shows the advantages, disadvantages, and characteristics of these methods. As presented in the table, sensor driver systems are cheap, but most of them are unavailable in bad weather or at night because the sensors directly depend on sunlight and temperature. By contrast, microprocessor driver systems are more effective However, they require a perfect tracking algorithm to efficiently move the photovoltaic modules. A combination of both techniques can increase the gain. Open–closed loop drivers work properly and increase the system gain. Moreover, using intelligent driver systems can determine MPP, and intelligent drivers can solve numerous problems in the tracking process of the sun, especially in changing weather situations.

The interest in driving tracking systems has increased nowadays compared to the past. Studies have proven that using driving systems increases the gained power compared with using fixed panels. However, current studies are focusing on how to track the position of the sun efficiently to increase the gained power rather than finding MPP. Several studies have focused on designing and improving solar tracking systems. Table 2 shows several studies that discussed and proposed different technologies to drive solar tracking systems conducted since 2005. Fig. 25 shows the number of categorized papers given in Table 2 according to their publication years. Fig. 25 proves that focusing on proving driving methods increased in 2017 compared with that in the past. We can expect more interest in the future, and more studies will aim to improve the current used technologies because of the high investment in solar energy nowadays.

The number of categorized papers according to the technology used to drive solar tracking systems is shown in Fig. 26. As shown in Fig. 26, the interest in using intelligent driver systems has increased compared with using other technologies. This is because these intelligent systems have several characteristics and AI principles that allow them to behave

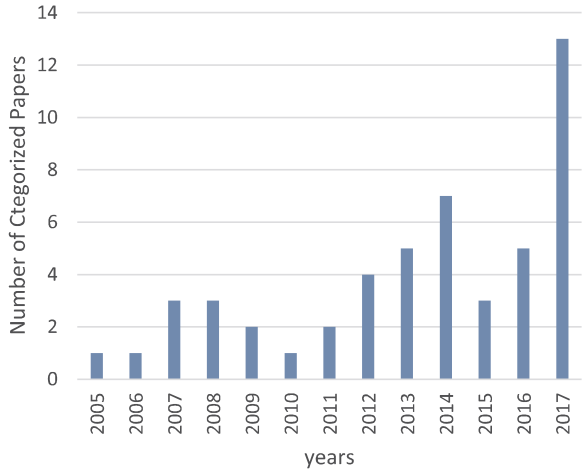


Fig. 25. Number of categorized papers according to published years.

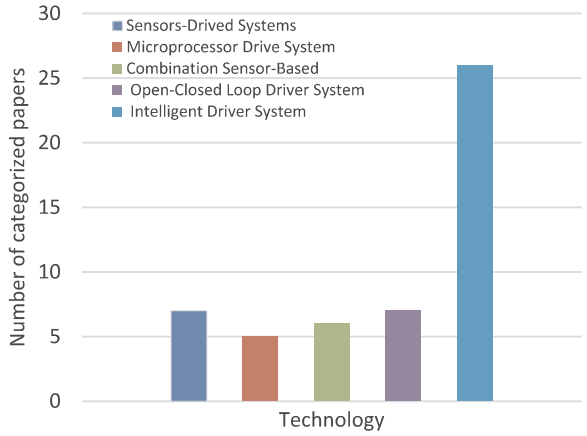


Fig. 26. Number of categorized papers according to technology.

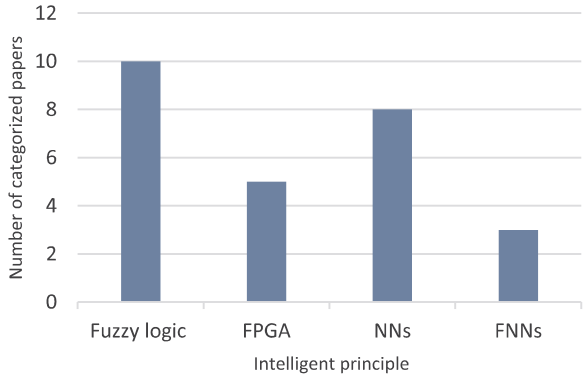


Fig. 27. Number of categorized papers based on intelligent derived systems.

like humans. Moreover, using a fuzzy logic controller in intelligent driver systems is common, although it suffers from many problems. A few studies have focused on using fuzzy neural network driver systems. Fig. 27 shows the number of categorized papers based on the intelligent principles used.

5. Limitations and future trends of solar tracking systems

Solar tracking systems have very high efficiency and performance compared with fixed or stationary solar photovoltaic systems. The main advantage of solar tracking systems is the increased electricity generation depending on the geographical location of the solar tracker and other variables. However, solar tracking systems possess numerous limitations. Solar tracking systems are more expensive than fixed systems due to the complexity of the technology used and their use of expensive products for their complex operations. These increased costs can be attributed to the need for frequent maintenance to ensure the proper functioning of these systems. Installing a solar tracking system can add further limitations to such systems. The installation process requires experts because of the necessary measurements and calculations to make these systems working properly. Moreover, the cost/performance of the solar tracking systems is not fixed for all types of trackers because numerous variables, such as the weather, the position of the sun in the sky, the country, and the type of solar tracker system itself, must be considered. The characteristics of the photovoltaic module are an important limitation because a few negative effects, such as dust and black spots, can emarge based on these characteristics.

Improving the technology of solar tracking systems is an important goal today and in the future, given the high demand for energy resources. Numerous researchers are attempting to change the future of solar tracking systems and eliminate current limitations. The idea is to increase the usability of solar energy in all sectors of life and introduce a replacement for petroleum energy. Changing the manufacturing characteristics of solar photovoltaics can affect the performance of solar tracking systems. However, the characteristics of solar photovoltaics are not enough to achieve the promised goal. Therefore, numerous novel methods have been proposed to enhance solar tracking systems. Controlling the movement of solar photovoltaics is the main idea to control the daily gained average solar energy. Solar tracking systems are promising resources of energy for the future because of their characteristics and advantages over current energy resources.

With regard to the future of solar tracking systems, the focus should be on adding new methods to efficiently track the sun. Numerous methods, such as using expert systems or intelligent techniques to control solar photovoltaic module, can be proposed or improved to achieve maximum usage of solar energy. Numerous intelligent techniques, such as neural networks, fuzzy logic, genetic algorithm, particle swarm optimization, can also be enhanced to make them suitable for the future use of solar energy.

6. Conclusion

The main purpose of this work was to present and categorize diverse types of solar tracking systems based on their technologies and driving methods. Solar tracking systems can be divided into two main types, namely, active and passive solar tracking systems. Active solar tracking systems use gears and motors to control photovoltaic modules, whereas passive tracking systems use a low-boiling-point compressed gas fluid that originates from solar heat. This work classified active solar tracking systems into five categories based on the driving methods employed. The five categories are sensor driver systems, microprocessor driver systems, open– closed loop driver systems, intelligent driver systems, and a combination of two or more of these driver systems. Intelligent driver systems are the most promising among these tracking systems due to their capability to predict the exact position of the sun by using on learning algorithms. Thus, most tracking systems currently focus on using intelligent principles. Meanwhile, many new efficient solar tracking systems will be designed and implemented using intelligent principles in the future.

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